

Flame Structures in a Model Solid Fuel Ramjet with Increasing Reynolds Numbers

Ryan DeBoskey*

NC State University, Raleigh, NC 27695

Syntek Technologies Inc., Severna Park, MD 21146

Trushant K. Patel,[†] David A. Kessler,[‡] Andrew M. Hess,[§] Brian T. Bojko,[¶] Ryan F. Johnson,^{||} Gabriel B. Goodwin,^{**}
U.S. Naval Research Laboratory, Washington, DC 20170

Venkateswaran Narayanaswamy,^{††}

NC State University, Raleigh, NC 27695

This paper takes a step towards understanding the complex nature of the combustion dynamics within a solid fuel ramjet operating at a variety of altitudes, outlining viable strategies for performing physically-accurate numerical simulations at reasonable computational costs. First, simulations of one-dimensional solid-fuel diffusion flames were performed to demonstrate the effects of scaling system pressure on flame structure. Next, a series of parametric 2D numerical simulations of a planar solid fuel ramjet with hydroxyl-terminated polybutadiene (HTPB) as fuel were performed. Grid convergence study established a relationship between upstream flow conditions and resolution requirements at the grain surface. Simulations with increasing operating pressures were performed to analyze the altitude effects of on flame dynamics. Proper Orthogonal Decomposition (POD) and Spectral Proper Orthogonal Decomposition (SPOD), flame brush probability density functions (PDF), flame curvature, and mean field analyses were used to compare the numerical solutions to each of the simulated conditions. The results showed that increasing the operating pressure (decreasing altitude) of the solid-fuel ramjet combustor caused an increase in fuel regression rate and flame curvature. However, other parameters such as the flame brush, flame temperature were insensitive to the operating pressure, which indicated combustor performance was largely self-throttling and

*National Defense Science and Engineering Graduate Fellow, PhD Student, Department of Mechanical and Aerospace Engineering, and AIAA Student Member

[†]Research Aerospace Engineer, Laboratory for Computational Physics and Fluid Dynamics, AIAA Member, and corresponding author: trushant.k.patel.civ@us.navy.mil

[‡]Director, Laboratories for Computational Physics and Fluid Dynamics, and AIAA Senior Member

[§]Research Mechanical Engineer, Laboratories for Computational Physics and Fluid Dynamics, and AIAA Member

[¶]Research Mechanical Engineer, Laboratories for Computational Physics and Fluid Dynamics, and AIAA Senior Member

^{||}Research Aerospace Engineer, Laboratories for Computational Physics, and Fluid Dynamics, and AIAA Member

^{**}Research Scientist, Hypersonic Aerodynamics and Propulsion Section, and AIAA Member

^{††}Professor, Department of Mechanical and Aerospace Engineering, and AIAA Associate Fellow

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dictated by the combustor geometry.

Nomenclature

α, β	=	power law fit parameters
Δh	=	grid spacing [μm]
Δt	=	time step [s]
κ	=	flame curvature [m^{-1}]
ρ	=	density [kg/m^3]
σ	=	meshing scale ratio [s]
ϕ	=	equivalence ratio
Ψ	=	flame edge
C	=	normalized computational cost
D	=	hydraulic diameter [m]
h	=	backward-facing step height [m]
L	=	length (fuel grain) [m]
$\dot{m}$	=	mass flow rate [kg/s]
p	=	pressure [kPa]
$\dot{r}$	=	regression rate [m/s]
Re	=	Reynolds number
St	=	Strouhal number
T	=	temperature [K]
V	=	velocity [m/s]
x	=	axial length [m]
y	=	wall-normal length [m]
Y	=	mass fraction

I. Introduction

Solid fuel ramjets (SFRJs) are a variant of the traditional ramjet engine, that use a high-density solid fuel for thrust generation. The dense solid fuel is embedded within the combustion chamber, which simplifies the engine fabrication and minimizes the engine empty mass by not requiring fuel tanks, valving, piping, allied instrumentation, and controls. For an SFRJ, the freestream air is compressed and supplied to the combustor chamber, which eliminates the need to carry onboard oxidizer and substantially reduces the fuel-weight fraction compared to solid fuel rockets. Combustion in an

SFRJ is typically sustained by a backward-facing step within the combustion chamber that generates a hot low-velocity recirculation region suitable for flame stabilization. However, even with the simple geometry and weight reduction, the scalability and current deployment of SFRJs is severely limited due to a lack of physical understanding of the complex aerothermochemical dynamics inside the combustor.

Over the past several decades, a variety of numerical studies have been undertaken to better characterize the combustion behavior in SFRJs. Netzer et al. laid the fundamentals for SFRJ modeling that used steady two-dimensional primitive variable simulations [1]. Extension of the primitive variable model allowed for analysis of the effects of radiation, three-dimensional, geometric configuration, and inlet conditions on the performance of SFRJ engines [2–4]. These additions enabled the simulations to predict variation of the regression along the fuel grain and affirmed the observed experimental trends. Following this work, Elands et al. [5] incorporated probability density functions to the primitive variable approach to better fit the experimental trends. Liou et al. [6] utilized this probability density function approach and injection of gaseous fuel in lieu of pyrolysis to predict flammability limits with improved agreement with experimental results. Recent SFRJ computations employed the Reynolds Averaged Navier Stokes (RANS) simulations that included finite rate chemistry and empirical pyrolysis models to simulate combustor performance. Musa et al. [7–9] has performed a series of unsteady RANS simulations to accompany experimental investigation of several parameters on global system performance. Their simulations utilized a series of heavily-reduced finite rate kinetic mechanisms to investigate the introduction of swirl, geometric configuration, and inlet thermodynamics on the regression of polyethylene fuel and thrust augmentation. These simulations used an empirical Arrhenius rate law at the surface of the solid fuel grain to model the pyrolysis mechanism outlined originally by Bianchi et al. [10] for use in the thermochemical erosion of rockets nozzles. Further studies considered the sensitivity of inlet conditions on flame stability [11] and the choice of pyrolysis model and chemistry closure fidelity on predicted flame structure [12]. Overall, SFRJ RANS efforts largely follow similar efforts previously undertaken for hybrid rocket configurations [13, 14]. While these studies have given some fundamental insights into the relative importance of the various physical processes that control fuel regression rates, RANS modeling is fundamentally unable to capture the unsteady dynamics of the underlying flame structures, which results in inaccurate prediction of the average flow reattachment, surface heat transfer, and flame brush [12].

High-fidelity numerical simulations offer the possibility of directly probing the response of the combustion behavior to changes in the flow conditions, material properties, and the sub-grid models used to describe each of the underlying physical processes in the flow. Large-eddy simulations (LES) offer greater fidelity over RANS simulations by modeling the small-scale turbulence and fully resolving the large-scale coherent turbulent structures. A series of LES simulations have been conducted for geometrically similar hybrid rocket configurations [15, 16] and premixed dump-combustor configurations [17, 18]. These simulations provide valuable insights into the fluid dynamics within cavity-stabilized non-premixed multiphase combustors; however, such high fidelity simulations are relatively scarce for SFRJ configurations [19–21]. With the absence of many high-fidelity numerical simulations of SFRJs, questions remain regarding how the

coupled dynamics of the highly unsteady flowfield, the complex chemical kinetics of the heavy hydrocarbon fuels, and the transient dynamics of heat transfer to the fuel grains and pyrolysis at the fluid-solid interface affect flame behavior at near-limit conditions where simplified modeling approaches are likely to be insufficient. Particularly, investigation at off-design conditions remains critically under-explored, which is necessary to help understand many of the pressing challenges regarding flammability and ignition which hinders widespread deployment of SFRJ technology [22, 23]. The utility of simplified modeling approaches at these critical limits (and under highly dynamic conditions) is challenging due to the reliance on gross physical simplifications and geometrically-dependent empirical model coefficients [24].

In this paper, we consider the impact of increasing pressure on the unsteady fluid dynamics and combustion dynamics within model solid fuel ramjets. From an operational point of view, increasing the operating pressure nominally corresponds to operation at decreasing altitudes during steady flight. The pressure of the combustor, p , is directly related to the Reynolds number, Re , and hence computational cost of the numerical simulation. For example, doubling the inflow pressure results in a roughly order of magnitude increase in computational cost, with the increase for a fixed domain size and time duration summarized by:

$$C \approx \left(\frac{p_2}{p_1} \right)^{N+1} \quad (1)$$

where C is the increase in computational cost, p_2/p_1 is the pressure ratio, N is the number of spatial dimensions, and the additional +1 accounts for the smaller integration time step required due to reduced grid size. Developing appropriate relationships between low-pressure and high-pressure operation of SFRJ engines potentially can decrease the improvement cycle of engine design by orders of magnitude, motivating the following parametric studies.

The rest of this paper is organized as follows. Theoretical pressure-scaling effects are first discussed using one-dimensional solid fuel counterflow flames. Subsequently, results for a series of LES that detail the time-resolved unsteady flame dynamics within a model 2D planar SFRJ combustor are discussed. First, the impact of pressure boundary condition implementation strategies using converging and converging-diverging aft-end nozzles on the fluid dynamics in the combustor and overall computational cost is described. Next, a grid convergence study is performed based on the upstream combustor conditions to determine the resolution requirements at the solid-fuel grain. Finally, the effect of operating pressure on the combustion dynamics is discussed. Results for a series of parametric studies are presented and conclusions are drawn, highlighting the major trends in SFRJ flame dynamics with combustor scale and operating pressure.

II. One-Dimensional Theory

For a given geometry and fuel type, the overall rate of solid fuel regression in SFRJs depends mainly on the combustor pressure, fuel-port air mass flux, and inlet air total temperature. The empirically determined regression rate

typically takes the following form [25]:

$$\dot{r} = ap^\alpha G^\beta T_t^\gamma \quad (2)$$

where a , p , G , and T_t are the scaling factor, combustor pressure, fuel-port air mass flux, and inlet air total temperature, respectively. The power law model fitting parameters α , β , and γ are determined empirically by measuring regression rate through a series of parametric runs. Seminal work by Schulte [25] showed that the power model fit coefficients varied up to 50% depending on geometric size of SFRJ engine. However, all fits, regardless of fuel composition or combustor geometry, show similar trends with power model fits of α , β , and γ each centered around 0.35 to 0.5 [26, 27].

It should be reiterated that during steady flight, the changes in the combustor pressure are primarily caused by the changes in the freestream pressure (Notably, combustor pressure is also affected by combustion heat release and the mass flow of fuel due to solid fuel regression). Furthermore in typical flight altitudes, the freestream pressure is directly proportional to the freestream density and the mass flux G . Therefore, the regression rate power law for Re scaling through pressure should trend as of $\alpha + \beta$ due to the coupled effect.

To evaluate the above argument on the pressure scaling of the regression rates, a series of 1D flames were simulated in CANTERA using the Counterflow Diffusion Flame (CDF) solver [28]. A fully coupled solid-fuel boundary condition was used to define the fuel mass flux, pyrolysis gases, and surface temperature of the HTPB fuel grain. Additional details regarding the solver formulation, interfacial boundary condition specification, and HTPB thermophysical and kinetic rate parameters are provided in the Appendix. The model has been previously validated against experiment, demonstrating good agreement in the prediction of regression rate in opposed-flow solid fuel flames [29, 30]. A detailed 94-species, 614-reaction, high-temperature kinetics mechanism for 1,3-butadiene (C_4H_6) combustion developed by Laskin, Wang, and Law [31] was used to define the gas-phase reactions. The oxidizer inlet was separated from the solid-fuel boundary by 10 mm. For all simulations, air at 650 K and velocity of 2 m/s was impinged on the solid-fuel surface. This results in all simulations having the same global strain rate $K = v_o/L = 200$ 1/s, but with increasing pressure. Pressure was increased from 67.5 kPa to 540 kPa, representing 12.5% and 100% of the model SFRJ combustor pressure [32] considered in this work, respectively. Flame thickness was measured using a temperature threshold, which was developed to agree with experimental flame thickness measurement made using shadowgraph imaging [33]. Following this work, the flame thickness was defined as the distance between the fuel grain surface ($z = 0$) and the location where $T = 683$ K on the oxidizer side of the flame. $T = 683$ K represents a 5% increase from the inflow oxidizer temperature, $T_{ox} = 650$ K, used for these simulations.

Figure 1 shows the variation of temperature and flame structure with increasing pressure, illustrating the decrease in flame thickness. There is a steady rise in peak flame temperature with increasing pressure, with a nearly 10% increase in peak flame temperature from 67.5 kPa to 540 kPa. However, the flame thickness shows a nearly 300% reduction in

flame thickness from 67.5 kPa to 540 kPa, indicating its much more pronounced dependence on pressure compared to maximum flame temperature. Thicker flames have been shown to be less sensitive to small eddies, decreasing the level of turbulence induced flame wrinkling [34]. Translating to SFRJ combustors, a considerable increase to the level of turbulence-induced flame-wrinkling is expected at elevated pressures accompanied by minor effects on peak flame temperature.

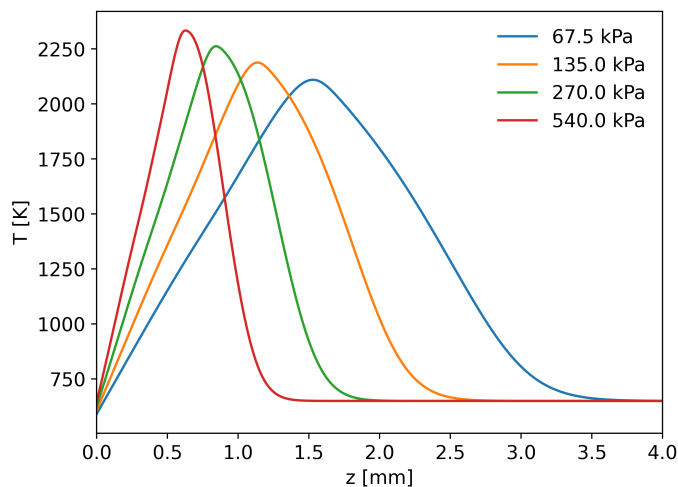


Figure 1 Temperature profiles computed for 1D solid fuel counterflow simulations at various operating pressures.

Figure 2 shows the variation of major flame-marking species mass fractions with increasing pressure. Similar to peak flame temperature, there are slight changes to the peak species mass fractions. The main effect of increasing pressure is increasing the gradient of species through the flame. The magnitude of the C_4H_6 mass fraction is relatively insensitive to operating pressure, with minor changes in peak mass fractions, whereas CO_2 , CO , and H_2O mass fractions increase by roughly 20-25% with increasing pressure. Extrapolating to SFRJ simulations, it is expected that the peak species mass fractions and species gradients near the wall will increase with increasing pressure. However, the average species and temperature distributions within the SFRJ combustor may remain relatively insensitive to variations in the Reynolds number (provided a similar flame brush with increasing pressure). Ideally, low pressure simulations may provide a good understanding of the outflow composition and overall combustion efficiency at high pressure, at significantly lower computational cost.

The full fuel surface composition (major species) for a representative 1D solid fuel counterflow flame at $p = 540$ kPa is provided in Table 1. The model HTPB is assumed to completely pyrolyze to C_4H_6 , resulting in the largest mass fraction at the surface. The rest of the surface composition is a direct result of diffusion to the solid fuel surface (via Equation 39). There is a large mass fraction of inert N_2 , which diffuses to the fuel rich side of the flame. Significant mass fractions of H_2O , CO , and CO_2 are observed, as well as intermediate fuel species that include H_2 , C_2H_2 , and C_2H_4 . The total sum of major species is 0.978.

Table 1 Fuel surface composition for a representative 1D solid fuel counterflow flame at $p = 540$ kPa (major species: $Y_i > 0.01$)

Species	Y_i	ΣY_i
C_4H_6	0.417	0.417
N_2	0.344	0.761
H_2O	0.060	0.821
CO	0.049	0.869
CO_2	0.035	0.904
H_2	0.030	0.934
C_2H_2	0.027	0.961
C_2H_4	0.018	0.978

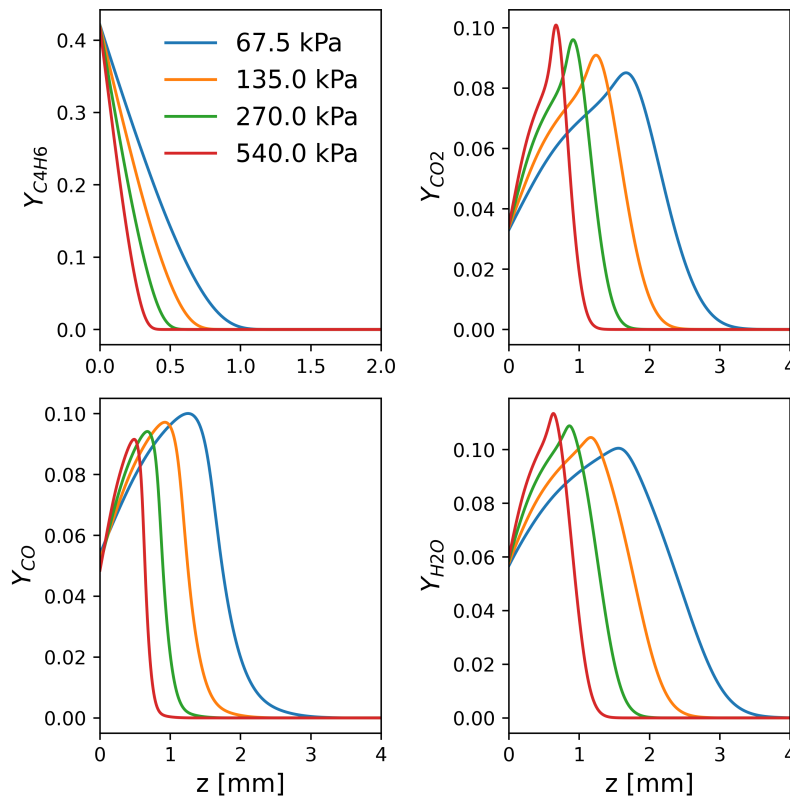


Figure 2 Mass fractions of major species profiles computed for 1D solid fuel counterflow simulations at various operating pressures.

Given the broad agreement in the expected trends of various thermochemical parameters, power law fits are modeled for regression rate and flame thickness using a simplified power law model cast in terms of pressure:

$$x = ap^{\alpha+\beta} \quad (3)$$

Table 2 Power law fits for regression rate and flame thickness computed for 1D solid fuel counterflow flames (p : [kPa])

Parameter	a	$\alpha + \beta$	R^2
Regression rate [mm/s]	$0.0032 \pm 4.e-5$	0.4809 ± 0.0020	0.99975
Flame thickness [mm]	24.37 ± 0.27	-0.4698 ± 0.0018	0.99980

where x is the parameter of interest. Figure 3 shows simulated regression rate and flame thickness with corresponding power law fits for flames at pressures from 100 kPa to 1000 kPa. Power law model fits are shown in Table 2 with corresponding R^2 values. There is excellent agreement between the modeled regression rate and flame thickness and the power law fits, with R^2 values greater than 0.99. From a computational cost perspective, high R^2 values of the power law fit imply that extrapolation of lower Reynolds number (Re) solution metrics to high Re solution is expected to be accurate in this simple configuration. Monotonic trends of species composition and profile structure further affirm reconstruction of higher Re solutions from lower Re solutions can be achieved.

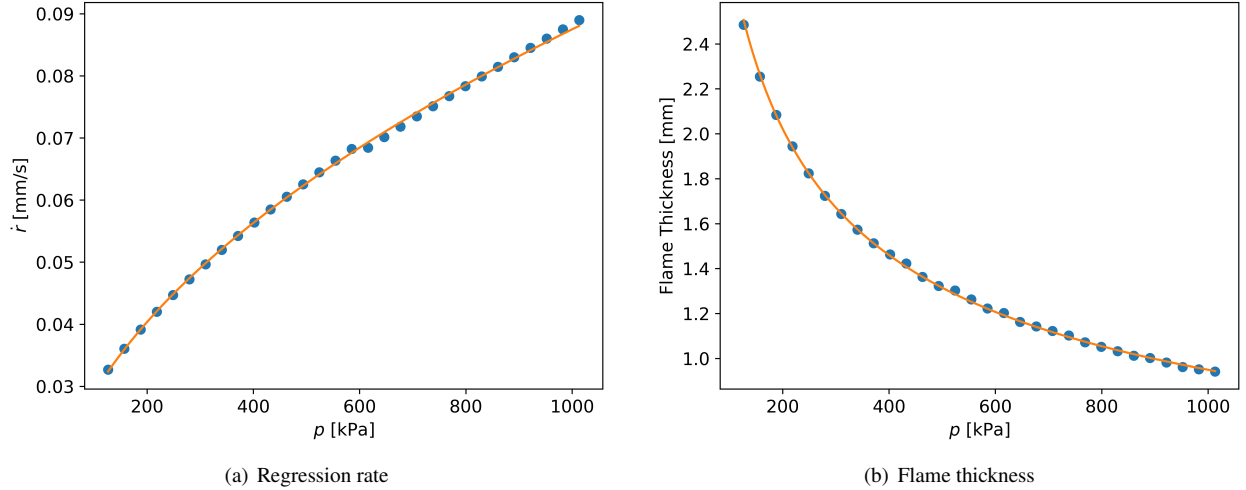


Figure 3 (a) Regression rate and (b) flame thickness computed for 1D solid fuel counterflow simulations (lines: Reynolds-number-scaling power law models, circles: simulation data).

III. Numerical Method

A. Governing Equations

Inside the fluid domain, we solve the compressible, multicomponent, chemically reacting Navier-Stokes equations, given generally by:

$$\frac{\partial y}{\partial t} + \nabla \cdot \mathcal{F}(y, \nabla y) - \mathcal{S}(y) = 0 \quad (4)$$

where y is the state vector, ∇y is its spatial gradient, t is time, $\mathcal{F}$ is the flux, and $\mathcal{S} = (0, \dots, 0, 0, \omega_1, \dots, \omega_{n_s})^T$ is the chemical source term, with ω_i corresponding to the net chemical production rate of the i th species. The physical coordinates are denoted by $x = (x_1, \dots, x_d)$, with d spatial dimensions. The vector of state variables is expanded as:

$$y = (\rho v_1, \dots, \rho v_d, \rho e_t, C_1, \dots, C_{n_s})^T \quad (5)$$

where ρ is density, $v = (v_1, \dots, v_d)$ is the velocity vector, e_t is the specific total energy, $C = (C_1, \dots, C_{n_s})$ is the vector of molar concentrations, and n_s is the number of species. The partial density of the i th species is defined as:

$$\rho_i = W_i C_i \quad (6)$$

where W_i is the molecular weight of the i th species, from which the density can be computed as:

$$\rho = \sum_{i=1}^{n_s} \rho_i \quad (7)$$

The mole and mass fractions of the i th species, X_i and Y_i , respectively, are given by:

$$X_i = \frac{C_i}{\sum_{i=1}^{n_s} C_i}, \quad Y_i = \frac{\rho_i}{\rho}. \quad (8)$$

The equation of state for the mixture is written as:

$$P = R^0 T \sum_{i=1}^{n_s} C_i, \quad (9)$$

where P is the pressure, T is the temperature, and R^0 is the universal gas constant. The specific total energy is the sum of the mixture-averaged specific internal energy, u , and the specific kinetic energy, written as:

$$e_t = u + \frac{1}{2} \sum_{k=1}^d v_k v_k \quad (10)$$

where the former is the mass-weighted sum of the specific internal energies of each species, given by:

$$u = \sum_{i=1}^{n_s} Y_i u_i \quad (11)$$

With the thermally perfect gas model, u_i is defined as:

$$u_i = h_i - R_i T = h_{\text{ref},i} + \int_{T_{\text{ref}}}^T c_{p,i}(\tau) d\tau - R_i T \quad (12)$$

where h_i is the specific enthalpy of the i th species, $R_i = R^0/W_i$, T_{ref} is the reference temperature of 298.15 K, $h_{\text{ref},i}$ is the reference-state species formation enthalpy, and $c_{p,i}$ is the specific heat at constant pressure of the i th species, which is approximated with a polynomial as a function of temperature based on the NASA coefficients [35, 36]:

$$c_{p,i} = \sum_{k=0}^{n_p} a_{ik} T^k. \quad (13)$$

The flux can be expressed as the difference between the convective flux, $\mathcal{F}^c$, and the viscous flux, $\mathcal{F}^v$:

$$\mathcal{F}(y, \nabla y) = (\mathcal{F}^c(y) - \mathcal{F}^v(y, \nabla y)), \quad (14)$$

where the k th spatial components are defined as:

$$\mathcal{F}_k^c(y) = (\rho v_k v_1 + P \delta_{k1}, \dots, \rho v_k v_d + P \delta_{kd}, v_k (\rho e_t + P), v_k C_1, \dots, v_k C_{n_s})^T \quad (15)$$

$$\mathcal{F}_k^v(y, \nabla y) = \left(\tau_{1k}, \dots, \tau_{dk}, \sum_{j=1}^d \tau_{kj} v_j + \sum_{i=1}^{n_s} W_i C_i h_i V_{ik} - q_k, C_1 V_{1k}, \dots, C_{n_s} V_{n_s k} \right)^T \quad (16)$$

τ is the viscous stress tensor, q is the heat flux, and V_{ik} is the k th spatial component of the diffusion velocity of the i th species, defined as:

$$V_{ik} = \hat{V}_{ik} - \frac{\sum_{l=1}^{n_s} W_l C_l \hat{V}_{lk}}{\rho}, \quad \hat{V}_{ik} = \frac{\bar{D}_i}{C_i} \frac{\partial C_i}{\partial x_k} - \frac{\bar{D}_i}{\rho} \frac{\partial \rho}{\partial x_k}, \quad (17)$$

which includes a standard correction to ensure mass conservation (i.e., $\sum_{i=1}^{n_s} W_i C_i V_{ik} = 0$) [37, 38]. $\bar{D}_i$ is the mixture-averaged diffusion coefficient of the i th species, obtained as [39]:

$$\bar{D}_i = \frac{1}{\bar{W}} \frac{\sum_{j=1, j \neq i}^{n_s} X_j W_j}{\sum_{j=1, j \neq i}^{n_s} X_j / D_{ij}}, \quad (18)$$

where $P_{\text{atm}} = 1$ atm, $\bar{W} = \rho / \sum_i C_i$ is the mixture molecular weight, and D_{ij} is the binary diffusion coefficient between the i th and j th species, which is a positive function of temperature and pressure [28, 40]. Note that $\bar{D}_i$ can be nonzero for $C_i = 0$. The k th spatial components of the viscous stress tensor and the heat flux are written as:

$$\tau_k(y, \nabla y) = \mu \left(\frac{\partial v_1}{\partial x_k} + \frac{\partial v_k}{\partial x_1} - \delta_{k1} \frac{2}{3} \sum_{j=1}^d \frac{\partial v_j}{\partial x_j}, \dots, \frac{\partial v_d}{\partial x_k} + \frac{\partial v_k}{\partial x_d} - \delta_{kd} \frac{2}{3} \sum_{j=1}^d \frac{\partial v_j}{\partial x_j} \right) \quad (19)$$

where μ is the dynamic viscosity, calculated using the Wilke model [41], and:

$$q_k(y, \nabla y) = -\lambda_T \frac{\partial T}{\partial x_k} \quad (20)$$

where λ_T is the thermal conductivity, computed with the Mathur model [42], respectively.

Equations (4–16) were solved using an extended version of the JENRE[®] Multiphysics Framework [43] that has been modified to include the gas-solid coupling at the solid fuel surface. The underlying flow solver chosen for this work utilized a nodal discontinuous Galerkin finite-element formulation with two-dimensional DG discretization. This formulation allowed the use of high-order polynomial basis functions, which increased the formal order of accuracy of the spatial flux computations. For these calculations, the HLLC approximate Riemann solver was used to compute the numerical flux across adjacent element faces, and the Bassi-Rebay (BR2) method [44] was used to discretize the viscous fluxes. Time integration was done explicitly using a strong stability-preserving Runge-Kutta method. For the finite-rate kinetics, reaction source terms are applied using a Strang-splitting approach, in which the system of stiff chemical reaction rate laws is integrated implicitly using the DGODE method as described by Johnson & Kercher [43]. The spatial discretization error of the DG formulation had been verified in earlier work to converge optimally with the expected rate of $(p + 1)$ rate for multicomponent, chemically reacting, flows [43]. Solutions in this work utilize DG($p = 1$) and DG($p = 2$) elements, resulting in nominal 2nd- and 3rd-order spatial and temporal accuracy, respectively.

B. Pyrolysis Model

For this work, we utilize a simplified model for the pyrolysis of the solid fuel grain as described in previous work [21, 33]. The simplification is based on the assumption of thermal equilibrium with the solid fuel grain, such that there is negligible heat flux (and hence, temperature gradient) inside the solid material away from the pyrolysis zone. This assumption eliminates the need to solve for the energy transport in the solid fuel material and the material interface condition reduces to a boundary condition for the gas-phase flow that sets the temperature, density, and species mass fractions at the solid fuel grain according to the following set of conservation equations:

$$\frac{\dot{m}_s''}{W_{sg}} = C_{sg}u - D_{sg} \left(\frac{\partial C_{sg}}{\partial n} - \frac{C_{sg}}{\rho} \frac{\partial \rho}{\partial n} \right) \quad (21)$$

$$0 = C_i u - D_i \left(\frac{\partial C_i}{\partial n} - \frac{C_i}{\rho} \frac{\partial \rho}{\partial n} \right) \quad (22)$$

$$\dot{m}_s'' L = \lambda_f \left(\frac{\partial T}{\partial n} \right)_f \quad (23)$$

where the subscripts ‘s’, ‘sg’, and ‘f’ refer to the solid phase, gas-phase pyrolysis product species, and the fluid region, respectively. The equations enforce the conservation of mass, species, and energy across the interface, with the left- and right-hand sides representing the solid- and gas-phase, respectively. Derivatives are computed in the direction ‘ $\hat{n}$ ’ normal to the interface, and L is the temperature-dependent latent heat of vaporization of the solid species. The surface

blowing velocity is $u = \dot{m}_s'' / \rho$, where ρ is the density of the gas in the pyrolysis zone. Notably, recent work has shown that the inclusion of heat flux through the surface into the solid fuel has a negligible effect on flame standoff and a minor effect on computed regression rate ($\approx 5\%$) [30].

The mass flux of the pyrolysis product, $\dot{m}_s''$, is computed using a simplified model in which solid fuel is converted to a gaseous fuel species according to an Arrhenius law:

$$\dot{m}_s'' = \rho_s \dot{r} = \rho_s l_r A_p e^{-E_p/R^0 T_I} \quad (24)$$

where $\dot{m}_s''$ is the mass flux of fuel into the fluid region, l_r is the thickness of the pyrolysis zone, T_I is the temperature at the fluid-solid interface, and $\dot{r}$ is the fuel regression rate. Several experimental studies have interrogated the liberated species from HTPB fuel pyrolysis using infrared and mass spectroscopy, determining butadiene (C_4H_6) as the primary pyrolysis product [45–47]. Thus, we consider a solid fuel composed entirely of a hypothetical polymer with properties similar to hydroxyl-terminated polybutadiene (HTPB) that is converted entirely to the gaseous hydrocarbon, butadiene (C_4H_6). The thermophysical properties of the model fuel are identical to those used for modeling HTPB in the one-dimensional counterflow simulations, and may be found in the Appendix. The enthalpy of the solid fuel material is related to temperature via an equation of state, $h_s = C_p T_I + e_0$, where C_p is the specific heat of the material and e_0 is a reference energy that enforces consistency with the gas-phase equation of state for the corresponding vapor species. The temperature-dependent latent heat is then defined as the difference between the solid phase and gas-phase enthalpies, $L = h_s - h_g$. Equations (21–24) then define a set of $n_s + 2$ nonlinear equations, where Eqs. (21–22) gives the n_s species concentrations, Eq. (23) provides the interface temperature, and Eq. (23) defines the mass flux. The set of equations are solved using an iterative Gauss-Newton method.

C. Chemistry Closure

The net production of each species i , ω_i , is modeled using finite-rate chemistry:

$$\omega_i = \sum_{j=1}^{N_r} \nu_{ij} \left(k_{f_j} \prod_{k=1}^{N_s} C_k^{b'_{k,j}} - k_{r_j} \prod_{k=1}^{N_s} C_k^{b''_{k,j}} \right) \quad (25)$$

for a system of N_r chemical reactions with stoichiometric coefficients $b'_{k,j}$ and $b''_{k,j}$ associated with the j^{th} reaction and $\nu_{ij} = b'_{k,j} - b''_{k,j}$. We assume an Arrhenius form for each individual reaction where $k_{f_j} = A_j T^{B_j} e^{-E_{a,j}/RT}$ for the system of reactions with associated model coefficients A_j , B_j , and $E_{a,j}$, allowing also modifications for pressure fall-off and three-body reactions. For reversible reactions, the reverse reaction rates k_{r_j} are computed using the standard equilibrium constant approach [48].

This work utilized a pressure-dependent six-step global reaction mechanism calibrated for SFRJ conditions. A detailed description of the six-step mechanism and calibrated model coefficients is given in a previous work [21]. The

six-step mechanism was originally calibrated using the one-dimensional counterflow flame solver, demonstrating very good agreement in the prediction of regression rate, flame thickness, temperature, and species compositions with respect to detailed chemistry mechanisms [21, 31].

The reaction mechanism consists of 10 species: C_4H_6 , H_2O , CO_2 , CO , O_2 , OH , O , H , H_2 and inert N_2 . Reactions R3 and R4 consists of two breakdown pathways of 1,3-butadiene into intermediate CO and H_2 . Reaction R5 is the well-studied water-phase shift reaction, that serves as the only pathway for the complete oxidation of CO to CO_2 . R6 controls the rate of oxidation of H_2 into H_2O . Reactions R7 and R8 are recombination reactions that introduce radicals O , H and OH to mitigate temperature rise:



The six-step mechanism demonstrated a good balance between physical accuracy and computational cost in the simulation of SFRJs and solid-fuel counterflow configurations. The addition of several key flame-marking species, and dissociation reactions, that modulate peak temperature provide a better description of flame dynamics compared to simpler approaches. However, the mechanism slightly overpredicts temperature and captures a limited chemical space, notably missing endothermic fuel breakdown reactions.

D. Model Problem and Procedure

All numerical simulations were performed on a model two-dimensional planar SFRJ combustor, the design of which is based on a center-slice through the optically-accessible slab burner designed by Senior et al. [32, 49, 50]. In using a planar burner, there is a tradeoff between the detailed resolution of the turbulent flame dynamics and capture

of the engine performance, as the geometry investigated is not suited for real-world application and generation of thrust. Figure 4 shows a typical computational mesh utilized for the parametric studies. The orange line demarks the "combustor lipline" probe line used to analyze the reacting flowfield. Clean, heated air flows into the domain from the inlet surface and through a converging nozzle. The air enters the combustion chamber through a tapered inlet nozzle and encounters a backward facing step with height $h = 0.59$ in (15 mm). The solid fuel grain has length $l = 12.75$ in (324 mm). Downstream of the solid fuel grain, a section of inert (non-pyrolyzing) material, $l_i = 1.5$ in (38 mm), serves as a transition to the system exhaust. For the configuration shown in Figure 4, a subsonic nozzle is used in place of a choked supersonic outflow, allowing the pressure in the combustor to be controlled directly. The pressure at the combustor exit is prescribed to accelerate the flow to $M = 0.4$ at the exhaust and maintain the nominal operating pressure in the combustor. Alternative configurations more relevant to SFRJ performance that utilize a converging-diverging nozzle at the aft-end exhaust will also be considered and described in Section IV. The bottom surface is set as a slip wall to avoid the necessity of resolving the boundary layer that lies outside the region of interest for this study. For all cases, the inflow velocity is 93 m/s at 650 K, resulting in a Mach number of ≈ 0.2 at the combustor inlet. It is important to note, flow separation behavior and combustor dynamics are sensitive to inflow profile and turbulence specification. In this work, a flat profile was specified at the computational inflow, the location corresponding to the exit of a flow-conditioning section in experiment [49]. The tapered inlet nozzle was modeled as no-slip to provide realistic boundary layer and fluctuations at the combustor entrance, with enough length to accurately capture boundary layer development. We consider the nominal combustor pressure to be 540 kPa [32].

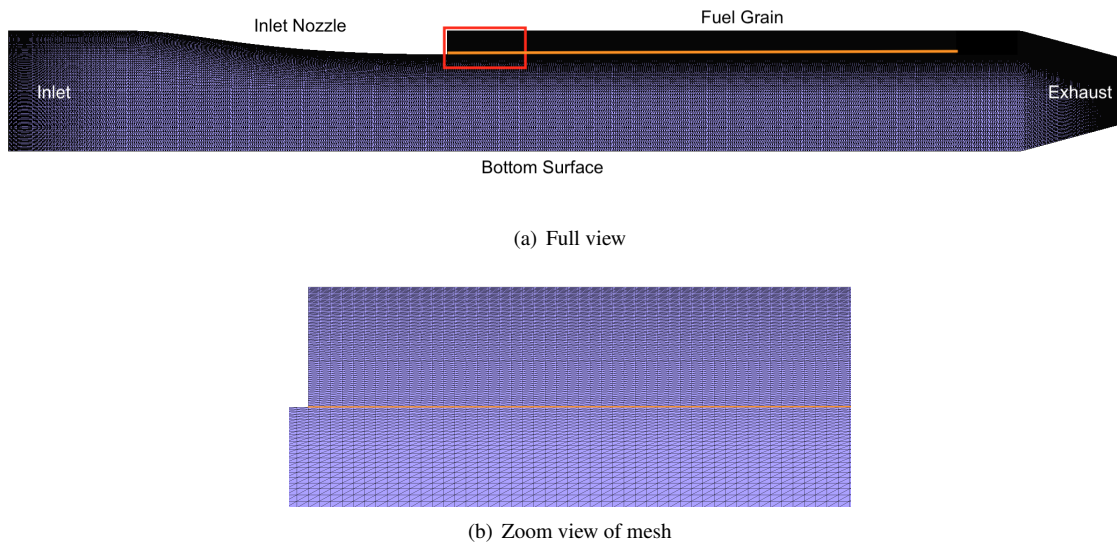


Figure 4 Prototypical computational mesh for model two-dimensional solid fuel ramjet combustor [49]. Structured mesh scaled based on grid resolution at the fuel grain surface

A structured mesh was used to keep consistency between the parametric runs, with element sizing throughout the

domain specified based on the freestream inlet conditions. It has been demonstrated experimentally that for a turbulent reacting flame anchored on a backward facing step, the Reynolds number in the shear layer is approximately an order of magnitude lower than the freestream Reynolds number as a result of increasing viscosity due to elevated combustion temperature [51]. Typically, wall-resolved LES require the general criteria of maintaining a near-wall grid-spacing such that $y^+ \approx 1$ for high-quality solutions accurately capturing surface gradients. Here, the relationship between freestream and the solid fuel grain surface near-wall grid spacing is taken to be:

$$\Delta s_{\infty, y^+=1} = \sigma \Delta s_{fuel, y^+=1} \quad (26)$$

where $\Delta s_{\infty, y^+=1}$ and $\Delta s_{fuel, y^+=1}$ is the resolution needed at freestream inlet and fuel grain, respectively, and σ is an unknown scaling factor. $\Delta s_{\infty, y^+=1}$ is calculated based on the methodology of White [52], using a reference length of 1 and a freestream velocity and temperature of 100 *m/s* and 640 *K*, respectively. The mesh was constructed given $\Delta s_{\infty, y^+=1}$ and σ , allowing for consistent mesh generation. A constant aspect ratio was used for the axial projection and cell size is increased from the top surface to the bottom surface, with cells in the shear layer and bottom surface 2× and 30× larger than the top surface elements, respectively.

A scaled nozzle parametric study was performed to assess the physical accuracy and cost savings of performing simulations of the SFRJ combustor without a choked outflow. The cases, listed in Table 3 are performed at a Reynolds number equivalent to the grid resolution parametric study by scaling the domain geometry by 1/8 and maintaining a pressure of 540 kPa. Results for these simulations were sampled for 10 flow through residence times, at a sampling rate of 200 kHz. These two cases do not consider a fully coupled pyrolysis model, C_4H_6 gas is blown into the domain with a mass flux of 0.1 *kg/m²s* and utilize a mesh scaling ratio of $\sigma = 5$.

Table 3 Description of scaled nozzle parametric study cases

Case	Scale	Pressure [kPa]	Re_L	Re_D	Re_h
Subsonic nozzle	1/8	540	367k	74.1k	17.0k
Supersonic nozzle	1/8	540	367k	74.1k	17.0k

Table 4 summarizes the grid resolution parametric study cases. The grid resolution study was performed at 1/8-scale pressure to conserve computational resources and determine the suitable mesh scaling ratio, σ . To provide context, the Reynolds number was calculated based on fuel grain length ($L = 324$ mm, step height ($h = 15$ mm), and hydraulic diameter ($D = 65$ mm) for each simulation. Results from both rectangular duct and backward facing step flows show that the Reynolds number considered is well above the critical number for exceeding the turbulent threshold [53, 54].

Previous work pertaining to turbulent reacting backward facing step flows suggests that the dominant Strouhal number for coherent vortex shedding is of the order of 0.1, $St = O(0.1)$, and best approximated by $St \approx 0.07$ [55].

Computed variables are sampled for 0.1 s at 10 kHz, resulting in 1000 snapshots of the time-evolving flowfield. This allows for an apriori estimation of over 50 vortex shedding frequency events at the dominant frequency. POD and SPOD were performed based on the methodology outlined by Weiss [56] and Schmidt [57], respectively, in order to capture and assess the unsteady dynamics within the SFRJ combustor. SPOD was performed using 8 blocks of 50 snapshots, resulting in a block overlap of 25 and resolution of 26 discrete frequencies. The flowfield snapshots contained 2500×571 samples in a two-dimensional rectangular plane starting at the backward facing step and extending to the end of the solid fuel grain.

Table 4 Description of grid resolution study cases

Mesh	Pressure [kPa]	σ	Δh_{min} [μm]	Re_L	Re_D	Re_h
Fine	67.5	3	62.1	367k	74.1k	17.0k
Medium	67.5	5	103.5	367k	74.1k	17.0k
Coarse	67.5	7	144.9	367k	74.1k	17.0k

Results from the grid resolution study (see Subsection IV.B) demonstrate the medium mesh is suitable of capturing the turbulent flame dynamics, and all other simulations were performed with a mesh scaling ratio of $\sigma = 5$. A Reynolds number parametric study was performed to assess the impact of pressure scaling on unsteady flame dynamics in the SFRJ combustor. The cases of interest are listed in Table 5, with simulations performed with inflow pressures of 67.5, 135, 270, and 540 kPa. Each simulation considered the same inlet temperature and velocity, resulting in a consistent Mach number between each run. The outflow pressure was scaled with the inflow pressure to maintain consistency in the nozzle expansion. Each simulation was run for two flow through residence times for statistical convergence before sampling begins. Results were sampled for a total of 6.7 ms at 150 kHz intervals, corresponding to two flow through residence times as defined by the $t_{res} = v_c/x_c$, where v_c and x_c are the combustor inlet velocity and combustor length, respectively.

Table 5 Description of Reynolds number parametric study cases. Grid scaling factor, $\sigma = 5$, determined from grid resolution study.

Pressure [kPa]	σ	Δh_{min} [μm]	Re_L	Re_D	Re_h	elements
67.5	5	103.5	367k	74.1k	17.0k	235k
135	5	51.8	735k	148k	34.0k	816k
270	5	25.9	1.47m	296k	68.0k	2.84m
540	5	12.9	2.94m	593k	136k	10.7m

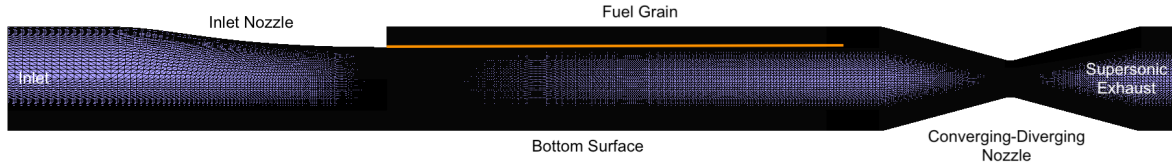


Figure 5 Computational mesh for supersonic nozzle simulation.

IV. Results

A. Supersonic and Subsonic Nozzle Comparison

Figure 5 shows the computational mesh with a converging-diverging supersonic nozzle. The advantage of using a supersonic nozzle is that the flow is always choked, and no knowledge of the back pressure is needed. The combustor pressure is determined based on the inlet mass flow, combustion process, and choked exhaust, providing a means for direct comparison with experiment. The disadvantage is the increase in computational cost due to the increased number of elements and time step limitations. For an explicit scheme the maximum time step, Δt , of the compressible Navier-Stokes Equations is proportional to [58]

$$\Delta t \propto \frac{\Delta h}{c + |v|} \quad (27)$$

where Δh is the local element size, c is the local speed of sound, and $|v|$ is the local magnitude of velocity. The flow is accelerated through the nozzle throat, resulting in a significant increase in the local velocity. This velocity increase combined with the existence of small local elements near the nozzle throat led to a considerable restriction in the maximum allowable time step. Use of a subsonic nozzle in lieu of a supersonic nozzle can help alleviate this restriction; however, the disadvantage is the difficulty associated with specification of the back pressure needed to achieve the desired mass flux, particularly during nozzle startup. Once steady combustion is reached, the back pressure at the exhaust is modulated until the combustor pressure matches the target experimental pressure. The time taken to iterate the combustor pressure is more than offset by the reduction in computational expense allowed by using a subsonic nozzle, resulting in significant speed-up.

A subset of computations addressed the possible discrepancies that can occur due to the choice of supersonic vs subsonic nozzle. Figure 6 shows a comparison of the combustor pressure and streamwise velocity, v_x , for the supersonic and subsonic nozzle geometries, where the dotted and solid lines refer to the combustor centerline and lipline, respectively. Both the supersonic and subsonic nozzle geometries show the same trends in pressure through the combustor. Along both the combustor lipline and centerline, there is a decrease in pressure due to the expansion behind the backward facing step followed by a sharp rise in pressure and a gradual decrease at the end of the combustor. The magnitude of the pressure between the simulations differ by 4 kPa ($< 1\%$ of total pressure). Notably, the recirculation

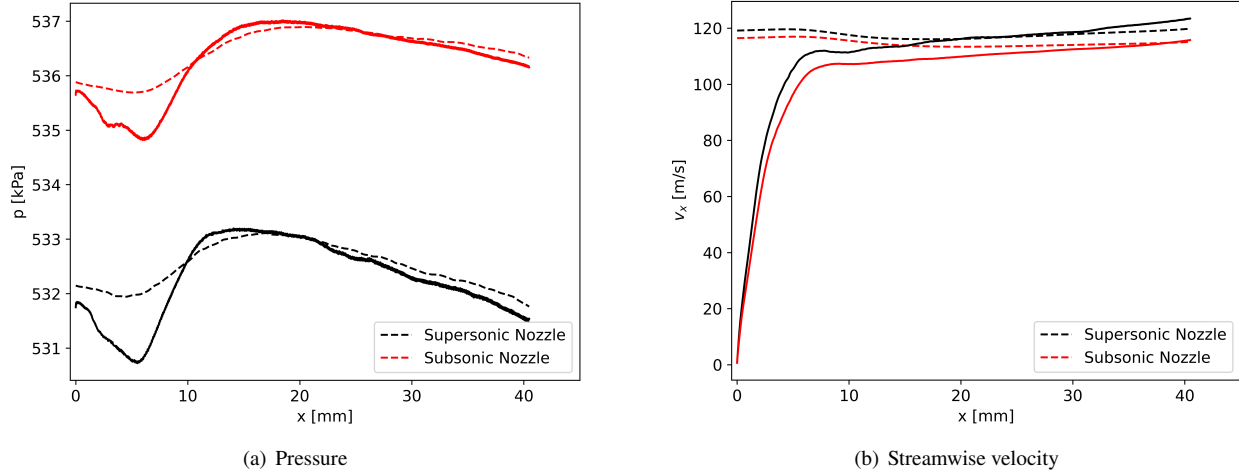


Figure 6 Comparison of (a) pressure and (b) streamwise velocity computed for the supersonic nozzle and subsonic nozzle combustor simulations (dotted: combustor centerline, solid: combustor lipline).

zone length may be inferred based on the adverse pressure gradient, with maximum pressure corresponding to the point of reattachment (observed at roughly $x = 15$ mm for both cases). The streamwise velocity shows similar trend-wise agreement. Along the combustor lipline, there is an increase in velocity through the shear layer that extends throughout the axial distance of the combustor. The centerline velocity decreases slightly due to the expansion at the step and then rises due to the combustion. As a result of slightly lower pressure in the supersonic nozzle case, there is a corresponding increase in velocity at both the combustor lipline and centerline. The results show excellent trend-wise agreement, with error in pressure between the two nozzle choices within 1%. The streamwise velocity error is within 3% everywhere along the centerline and within 8% everywhere along the lipline, with error due primarily to small differences in pressure.

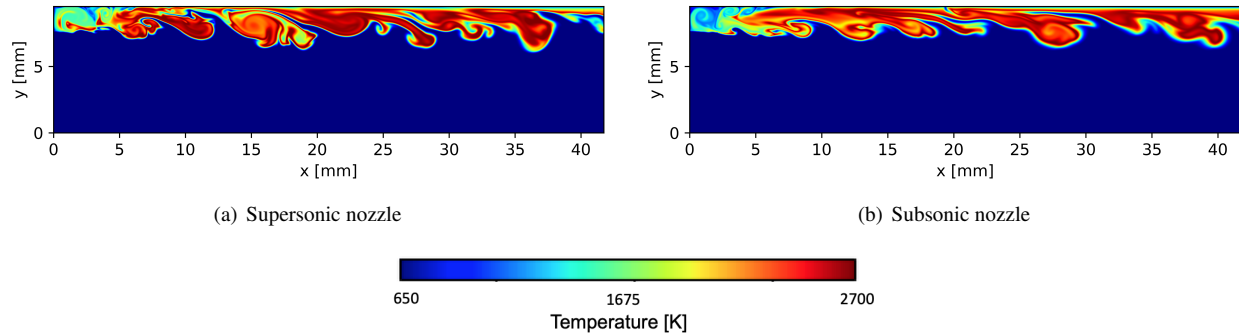


Figure 7 Comparison of instantaneous temperature fields computed for model SFRJ with (a) converging-diverging exhaust nozzle and (b) converging subsonic exhaust nozzle.

Figures 7 and 8 show 2D instantaneous and average temperature contours within the model SFRJ combustor, respectively. Both results show good qualitative and quantitative agreement between the two simulations. The maximum

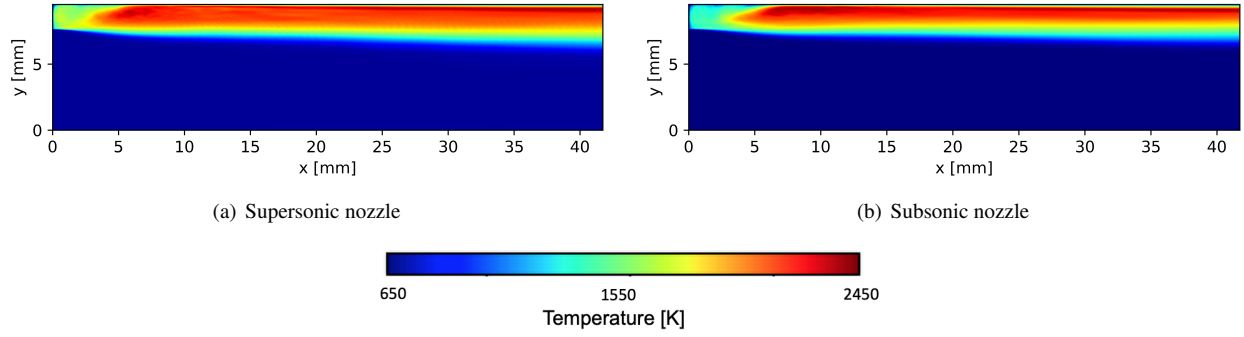


Figure 8 Comparison of temporally-averaged temperature fields computed for model SFRJ with (a) converging-diverging exhaust nozzle and (b) converging subsonic exhaust nozzle.

instantaneous temperature difference between the two instantaneous snapshots is within 10 K, and the maximum average temperature is within 5 K for the supersonic and subsonic nozzles. The average temperature contour is relatively insensitive between the two nozzle cases, with a slightly higher temperature seen in the recirculation region and a slightly more diffuse profile downstream seen for the supersonic nozzle. This can be attributed to the slightly higher velocity obtained, resulting in a more diffuse profile. There is again excellent trend-wise agreement between the two cases.

To isolate the effects of the differing nozzles, the pressure and streamwise velocity are normalized by the pressure and velocity at the entrance of the nozzle, P_{in} and $v_{x,in}$, respectively. Figure 9 shows the pressure ratio and streamwise velocity ratio along the centerlines of both nozzles, up to the termination of the subsonic nozzle. The results show excellent agreement, as both nozzles show nearly identical expansion, with slight deviation at the end of the subsonic nozzle.

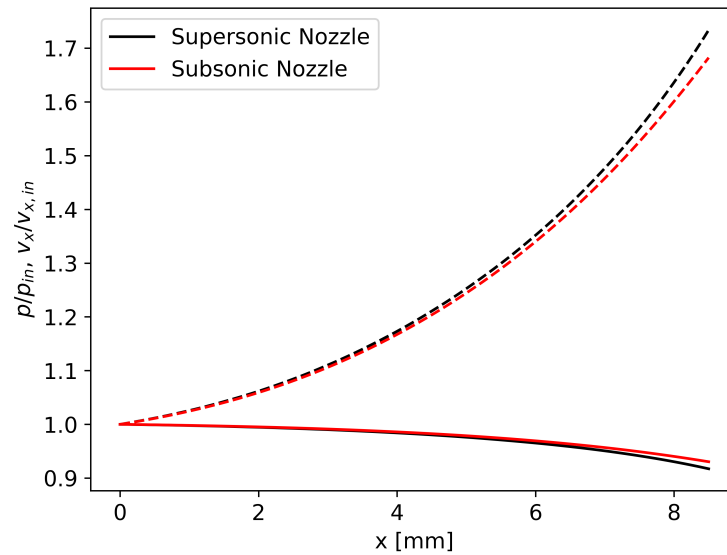


Figure 9 Comparison of $v_x/v_{x,in}$ (dotted) and p/p_{in} (solid) profiles computed for supersonic nozzle and subsonic nozzles.

Overall there are only minor differences in the observed pressure, velocity, and temperature within the model SFRJ for the cases with different exit nozzles. The expansion through the subsonic and supersonic nozzle is nearly identical and the difference in combustor performance can be attributed to the prescribed back pressure for the subsonic nozzle. Prescribing the back pressure requires an assumption of the pressure along the supersonic nozzle, resulting in small discrepancies through the combustor. However, the use of a subsonic nozzle despite the minor differences is justified due to the major associated computational savings and approximately $2\times$ speed-up; therefore, a subsonic exit nozzle was employed for the remainder of the simulations presented.

B. Mesh Refinement

Mesh refinement was performed on the 67.5 kPa pressure case to conserve computational cost, and to study the extension of the results to more computationally expensive simulations with higher pressure. First, the local y^+ along the fuel grain surface was calculated by

$$y^+ = \frac{h}{p+1} \sqrt{\frac{\rho}{\mu}} \sqrt{\frac{\partial u}{\partial y_n}} \quad (28)$$

where p is the polynomial order of the underlying basis function in the element and $\frac{\partial u}{\partial y_n}$ is the velocity gradient normal to the wall calculated using a second-order backward finite difference.

Figure 10 shows the local y^+ values across the solid fuel surface for simulations on the coarse, medium, fine meshes using DG ($p = 1$) elements and using DG ($p = 2$) elements on the coarse mesh. The results show decreasing y^+ magnitude with increasing mesh resolution, with the fine simulation having a maximum y^+ value of 0.75. The fine mesh maintains a y^+ below unity throughout the domain. The medium DG ($p = 1$) and coarse DG ($p = 2$) solutions have maximum y^+ values of approximately 1.1 within the recirculation region and below unity elsewhere. The coarse DG ($p = 1$) mesh has a y^+ value greater than 1.5. Capturing of the local y^+ along the solid fuel grain illuminates several important features critical to SFRJ performance. The minimum y^+ around $x = 0.05$ and $x = 0.15$ show the precise location of the predicted counter-rotating and main recirculation reattachment points, respectively. The smooth increase in y^+ downstream of the main reattachment point shows the transition to the diffusion dominated combustion regime. Results show that the medium and fine DG ($p = 1$) and the coarse DG ($p = 2$) solutions agree well in predicting the different regimes, with excellent agreement between the medium and fine domains in the prediction of the main reattachment point.

Figures 11-12 shows the instantaneous and average temperature contours for the coarse, medium, fine and coarse DG ($p = 2$) simulations, with $x = 0$ marking the backward facing step. Instantaneous plots show good agreement between all mesh resolutions in the prediction of maximum temperature and temperature within the recirculation region. Qualitatively, the coarse mesh shows much larger and smoothed out vortical structures compared to the finer meshes and

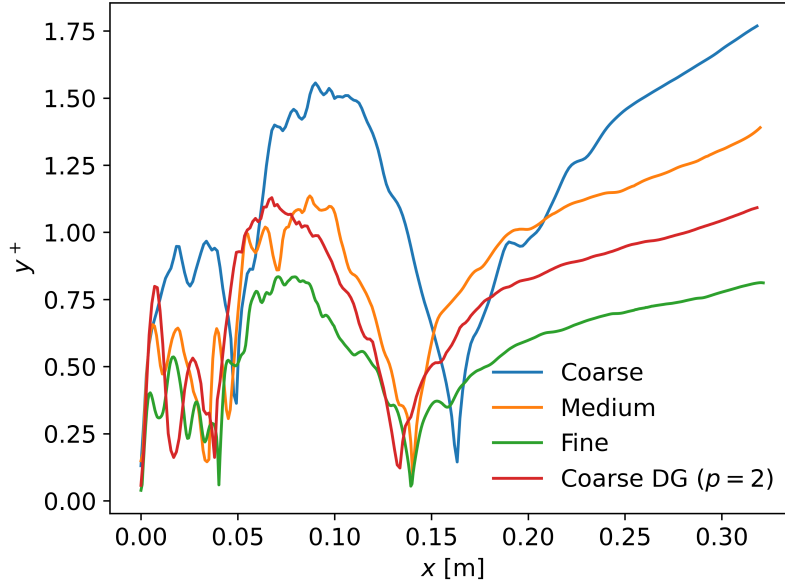


Figure 10 Variation in y^+ across fuel grain for mesh resolution parametric study.

the coarse DG ($p = 2$) mesh, indicating that the coarse DG ($p = 1$) mesh may be insufficient in capturing proper flame structure.

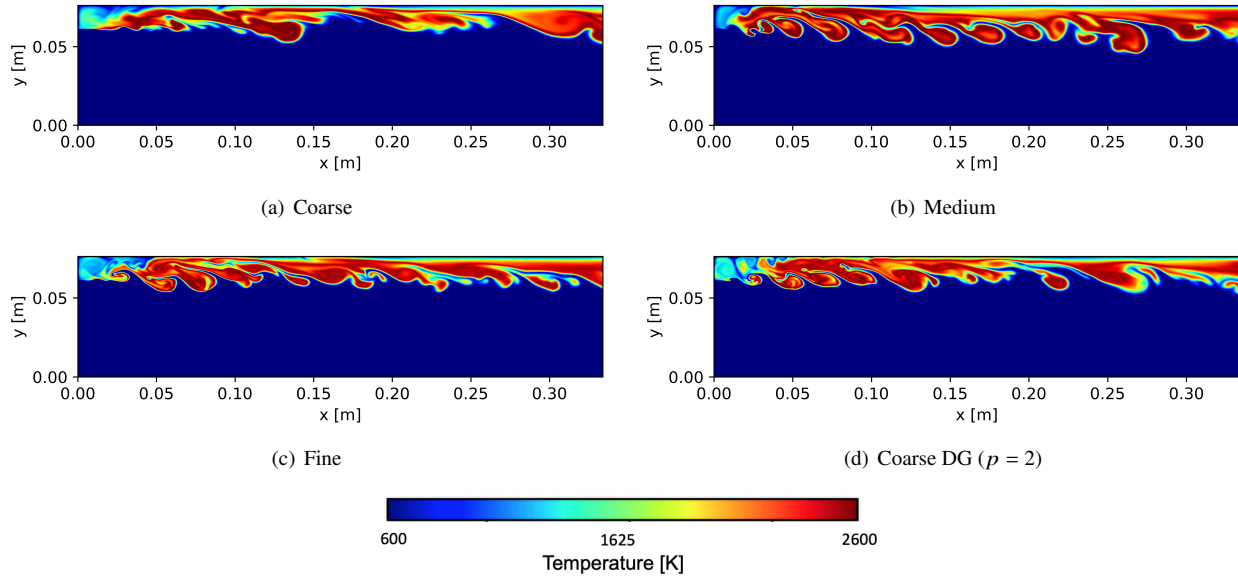


Figure 11 Instantaneous temperature fields computed on (a) coarse DG ($p = 1$), (b) medium DG ($p = 1$), (c) fine DG ($p = 1$), and (d) coarse DG ($p = 2$) meshes.

Average temperature plots show excellent agreement between all resolutions, indicating that all grid resolutions are useful for obtaining the average flowfield. There is a slightly larger high temperature region extending from the backward facing step lip in the coarse domain, which is not seen in the other refinement cases. All simulations predict

an average maximum temperature of 2300 K, and a temperature of around 1500 K in the recirculation region outside the main shear layer. The flame is anchored with the backward facing step, which draws in fresh air and limits the temperature in the recirculation region. The temperature is maximum near the fuel surface, indicating sharp gradients in temperature normal to the wall.

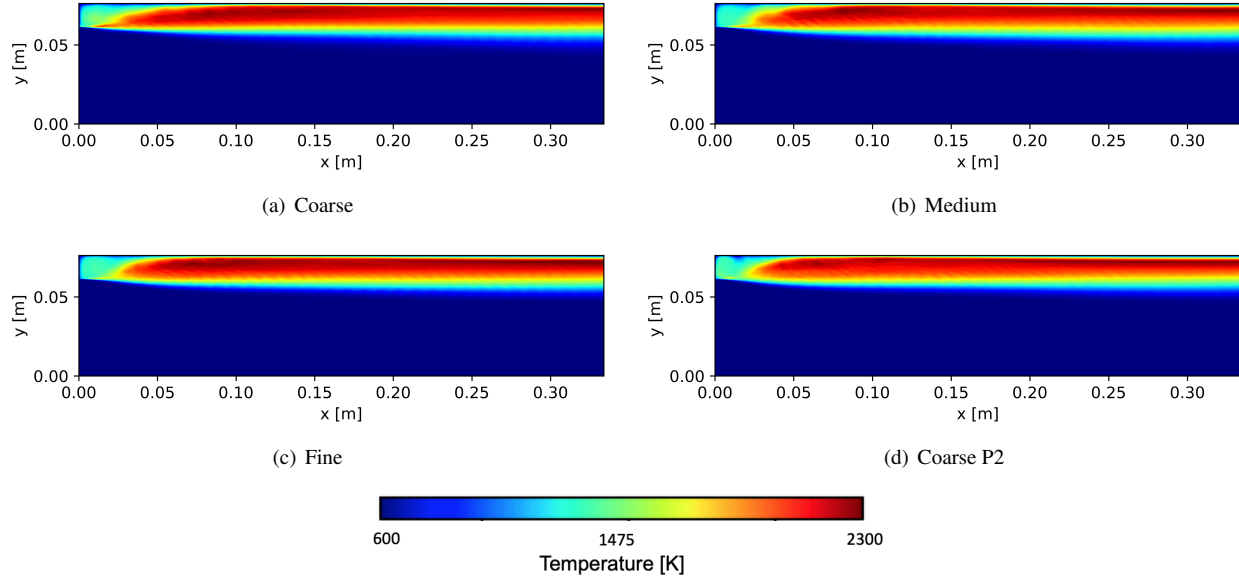


Figure 12 Average temperature fields computed on (a) coarse DG ($p = 1$), (b) medium DG ($p = 1$), (c) fine DG ($p = 1$), and (d) coarse DG ($p = 2$) meshes.

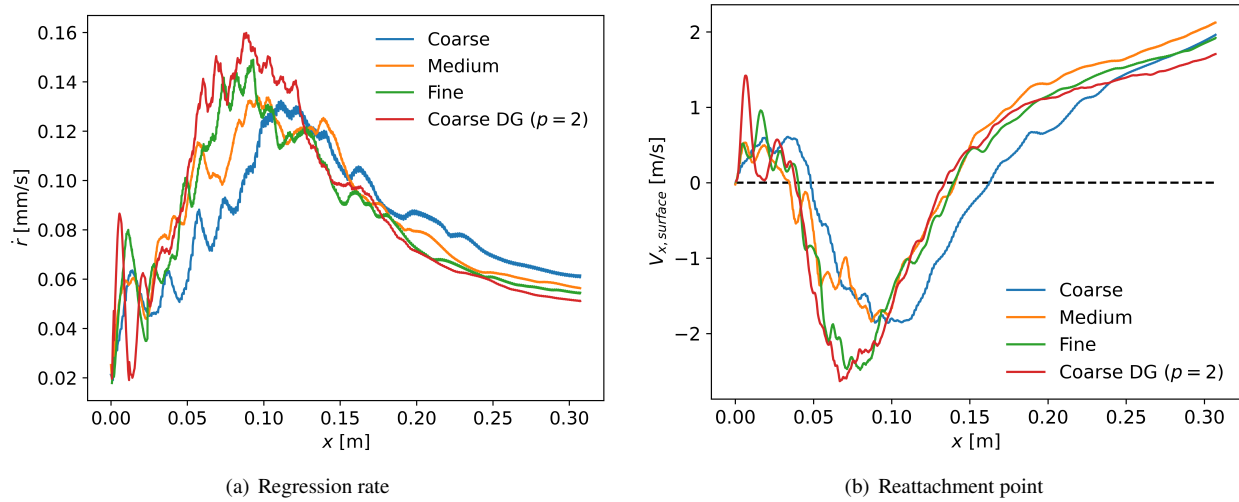


Figure 13 (a) Solid fuel regression rate and (b) streamwise velocity along the fuel grain computed using the various meshes used for the grid refinement study.

SFRJ performance is governed by the coupling of the turbulent heat transfer from the gas-phase combustion and solid fuel pyrolysis. Adequate resolution is required to capture the local temperature and velocity gradients at the solid

fuel surface in order to determine both the local regression rate and flow reattachment points. Figure 13 shows the local regression rate and streamwise velocity at the solid fuel surface for the mesh refinement cases. The medium, fine, and coarse DG ($p = 2$) domains show excellent agreement in the prediction of regression rate between $x = 0.03$ to $x = 0.07$ m and $x > 0.15$ m. Agreement in the peak regression rate magnitude is within 15 % between the three cases, with the medium, fine, and coarse DG ($p = 2$) mesh showing excellent agreement in the location of peak regression rate magnitude, $x \approx 0.09$ m. The maximum predicted regression is different between all cases, as a result of the non-linear Arrhenius regression rate model, Equation 24, in which small changes in the predicted solid fuel surface temperature result in large variations to the regression rate. The coarse resolution shows significant deviations in the predicted axial regression rate as compared to the finer simulations, indicating the resolution is not sufficient to capture the near wall flame dynamics and heat transfer.

Poor agreement of the coarse mesh is further evidenced by the prediction of the surface velocity, computed above the fuel grain surface and shown in Figure 13b. The medium, fine, and coarse DG ($p = 2$) domains show excellent agreement in the prediction of the main reattachment point, observed at $x \approx 0.13$ m, whereas the coarse mesh predicts the reattachment point significantly further downstream at $x = 0.16$ m. The solutions show evidence of a counter-rotating vortex at the backward facing step, with a smaller secondary reattachment point at $x \approx 0.05$ m. Again, the coarse mesh overpredicts the secondary reattachment point in comparison to the finer meshes. Of note, the magnitude of the maximum velocity of the medium mesh is slightly lower than the fine and coarse DG ($p = 2$) domain, showing some deviation from the finer solutions. Overall, the medium mesh is able to properly capture the main average reattachment points and main average pyrolysis trends in accordance with the fine and coarse DG ($p = 2$) cases indicating the efficacy of the medium mesh resolution. The coarse mesh is unsuitable for capturing either the near wall trends highlighting mesh dependencies.

While capturing of the average regression rate, reattachment point, and temperature within the model SFRJ is very important to engine performance, the ability to delineate the complex coupling between unsteady flow structures, heat transfer, and pyrolysis are critical to understanding the governing processes that control the engine performance. To this end, evaluation of the dominant time dependent structures from LES was performed using POD and SPOD to determine suitable resolution requirements. POD and SPOD was performed on the sampled instantaneous wall-normal velocity over 0.1s at 10 kHz. Figure 14 shows a summary of the POD and SPOD results for the mesh refinement cases. There is excellent agreement between the medium, fine, and coarse DG ($p = 2$) cases for POD energy distribution in the first 10 modes. The energy slowly decays with increasing mode with fairly uniform distribution of energy. The coarse case shows poorer agreement with the other cases, with a high energy percentage in the first couple modes and a much higher decay in energy. Figure 14b shows the distribution of SPOD eigenvalues across St for Mode 0 in the model SFRJ. The results show good agreement between the medium, fine, and coarse DG ($p = 2$) cases and more modest agreement with the coarse case. There is an excellent agreement between the medium and fine cases, with peaks at $St = 0.05$ and

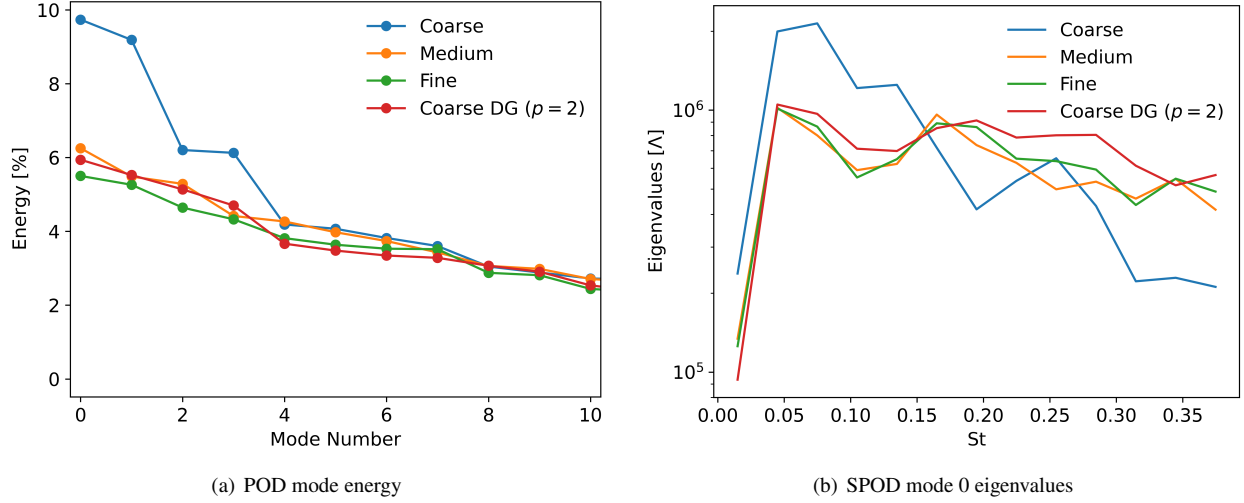


Figure 14 Summary of instantaneous wall-normal velocity a) POD mode energy and b) SPOD mode 0 eigenvalue frequency results for mesh refinement cases

$St = 0.15$. The coarse case shows a large peak at lower St numbers and a sharp decline as St number increases. The frequency distribution of the coarse case indicates that there are much larger coherent structures predicted by the LES in comparison to the other cases. This confirms quantitatively the qualitative results seen in the instantaneous temperature contours for the coarse domain. Results of the POD and SPOD calculations indicate that the medium mesh is capable of temporal resolution of the dominant turbulent structures within the model SFRJ domain.

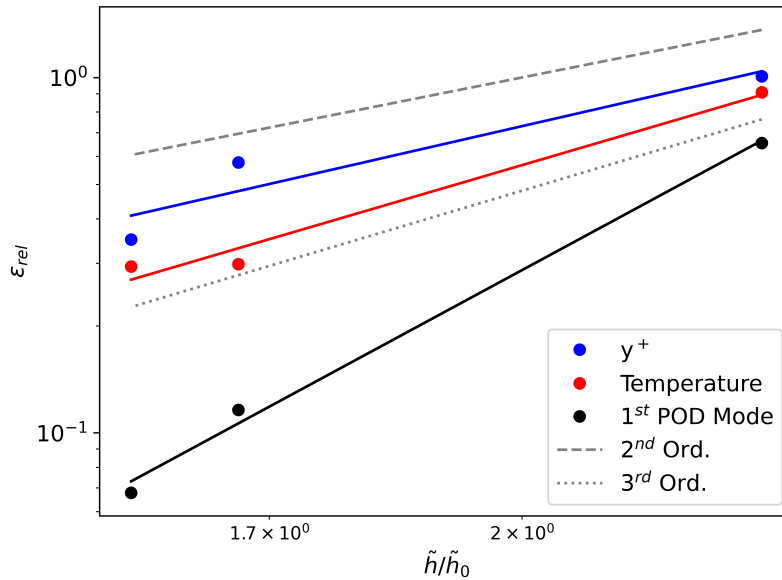


Figure 15 Quantitative grid refinement error analysis based on measured y^+ , centerline Temperature, and 1st POD mode energy

The mesh refinement results are quantified in Figure 15, which shows a log-log error analysis based on measured y^+ ,

centerline temperature, and 1st POD mode energy. The relative error, ϵ_{rel} is calculated as:

$$\epsilon_{rel} = \frac{||x - x_0||_{L2}}{||x_0||_{L2}} \quad (29)$$

where x and x_0 are the given solution quantity and fine solution used as a baseline truth, respectively. A normalized grid spacing, $\tilde{h} = \sqrt{DOF}$, is defined based on the square root of the number of degrees of freedom, DOF , to normalize between DG ($p = 1$) and DG ($p = 2$) simulations. The measured y^+ and centerline temperature relative errors converge between 2nd- and 3rd-order accuracy, as expected for the DG ($p = 1$) and DG ($p = 2$) scheme employed. However, the relative error for the 1st POD mode energy converges much faster, roughly 5th-order accurate, indicating that the size and energy of unsteady structures have a higher sensitivity to minor differences in grid resolution. Overall, the medium and coarse DG ($p = 2$) show similar levels of relative error, and are more accurate in comparison to the coarse case.

C. Pressure Scaling

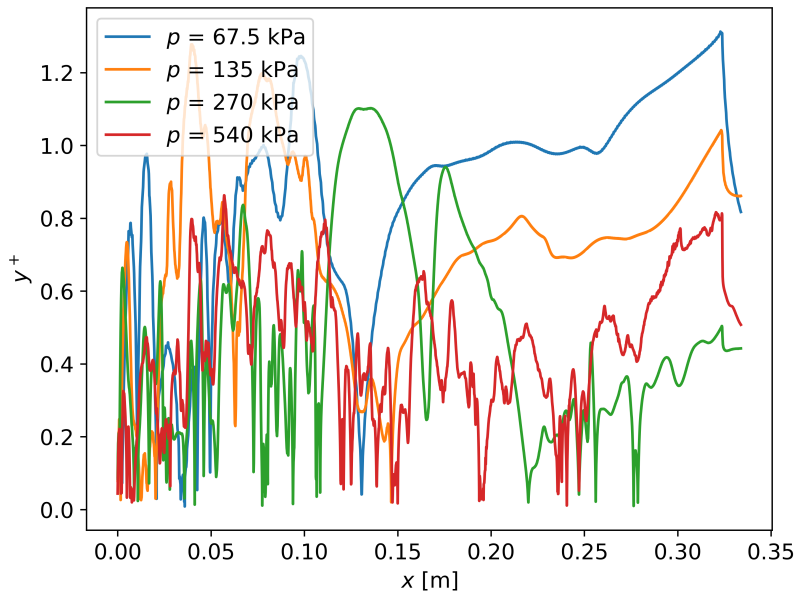


Figure 16 Variation of local y^+ along the fuel grain using the medium DG ($p = 1$) mesh for different operating pressure conditions.

Cumulative results of the mesh refinement study indicate that the medium DG ($p = 1$) mesh was suitable for capturing both the average and unsteady effects within the model SFRJ, and a mesh scaling factor of $\sigma = 5$ was used for all high pressure simulations. The average y^+ was sampled along the fuel grain surface to determine the suitability of performing mesh refinement on low-pressure simulations and development of mesh scaling based on upstream inflow conditions. Figure 16 shows the local y^+ value with increasing pressure. There is a slight trend with increasing pressure, showing a decrease in average y^+ along the fuel grain surface. The $p = 67.5$ and 130 kPa cases have maximum y^+ of

approximately 1.2, whereas the $p = 270$ kPa and 540 kPa have a maximum y^+ of 1.1 and 0.9, respectively. This confirms that the mesh resolution requirements for the pyrolyzing wall can be determined apriori based on the upstream flow conditions, and establishes $\sigma = 5$ a suitable mesh scaling factor for this model geometry. This mesh scaling factor will need to be modified for other SFRJ geometries, likely a function of step height and inlet Mach number. Performance of mesh refinement studies at lower pressure saves enormous computational cost and accelerates the pace of modeling improvement. Previous results from the mesh refinement yielded a new definition of the reattachment point based on the average y^+ profile. The $p = 270$ kPa and 540 kPa cases show considerably finer structures near the wall, requiring a longer sampling duration to quantify the average reattachment location

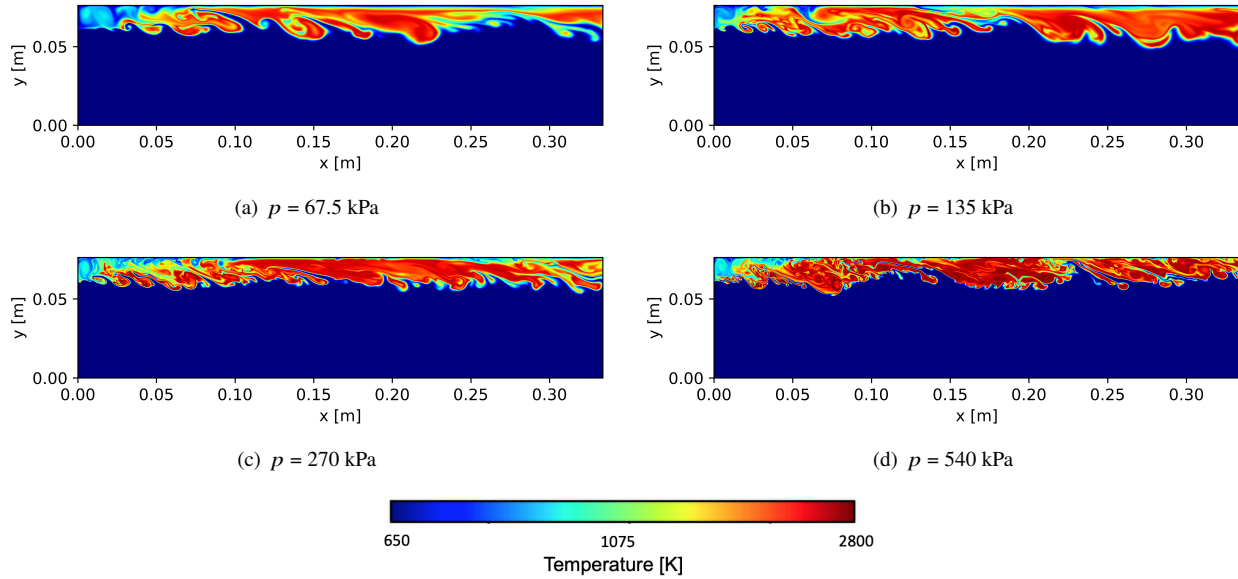


Figure 17 Instantaneous temperature fields near the fuel grain computed for increasing operating pressures using the medium DG ($p = 1$) mesh

Given adequate mesh resolution required to generate high-fidelity LES solutions, the effects of pressure scaling on the flame dynamics within the SFRJ combustor were explored. Figure 17 shows instantaneous temperature profiles within the model SFRJ combustor with increasing pressure. With increasing pressure there is a monotonic increase in maximum flame temperature and more flame induced wrinkling, consistent with one-dimensional analysis. With increasing pressure, the flame anchoring occurs closer to the backward-facing step lip indicating changes in flammability with pressure. At lower pressures there is a considerably larger standoff than in the high pressure cases, indicating a larger residence time is needed for reactions to progress. This is consistent with decreasing ignition delay times of 1,3-butadiene with increasing pressure [59], important for flame stability at different altitudes of operation. Particularly at the lowest pressure of $p = 67.5$ kPa, there is a much lower temperature in the recirculation zone and large flame standoff, indicating that the flame is nearing extinction limits. It is apparent that increasing pressure improves stability limits, as predicted by SFRJ flammability limit mapping [24], and care must be taken to ensure flammability is maintained before

mapping between high- and low-pressure operation.

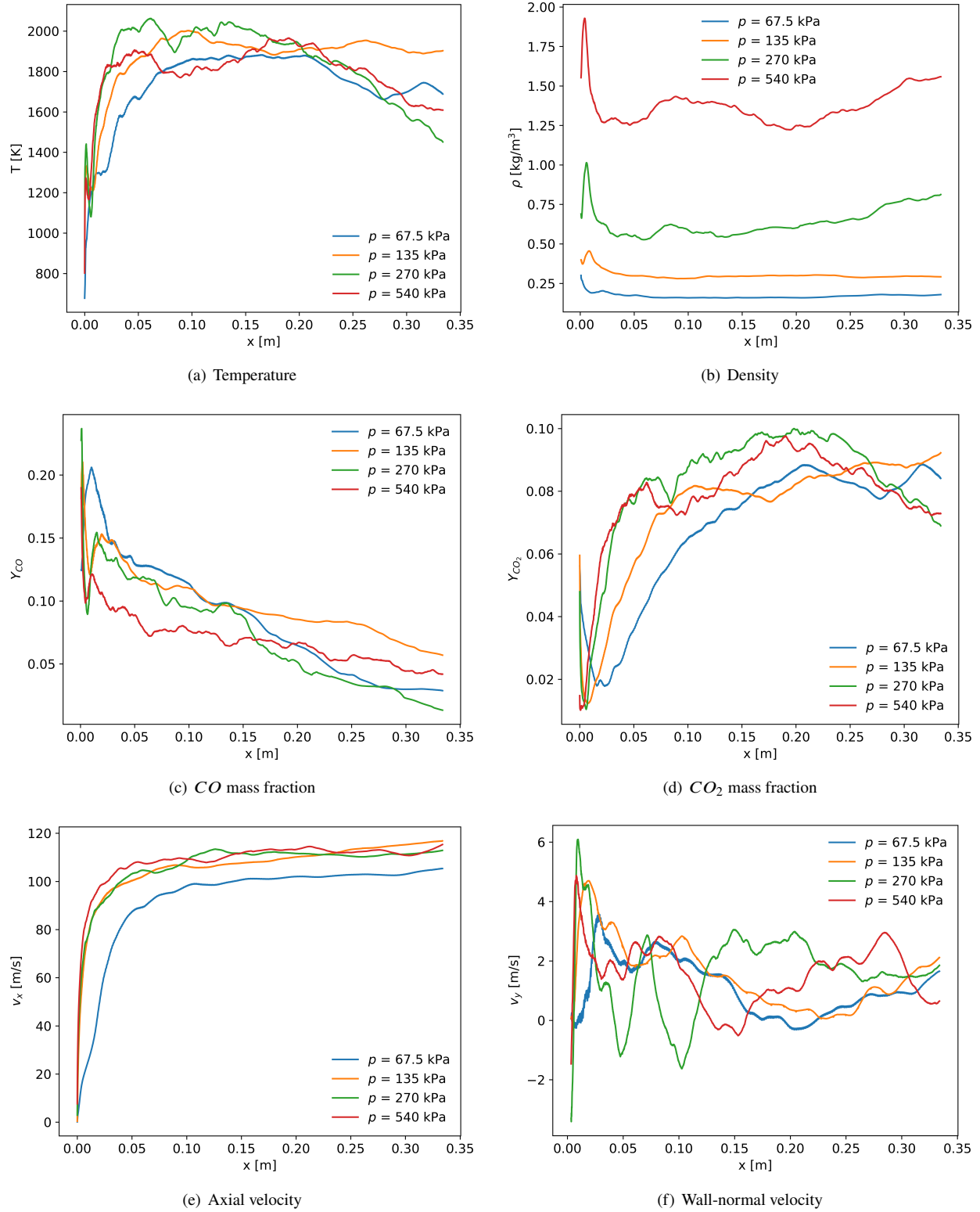


Figure 18 Comparison of lip-line profiles with increasing operating pressure

To further quantify the effects of pressure scaling, several average line samples from the flowfield were calculated. Due to the short sampling duration of only two flow through times, average trends that exhibit monotonicity indicate leading order effects whereas trends that do not show clear relationships with increasing pressure indicate pressure insensitivity or exhibit significantly weaker second-order relationships. These average plots exhibit significant uncertainty due to the short sample duration, as infrequent vortex shedding events are disproportionately exhibited between the different operating pressure simulations.

An average was taken along the combustor lipline, extending from the backward facing step axially through the combustor. Figure 18 shows a comparison of the average temperature, density, CO mass fraction, CO_2 mass fraction, axial velocity, and wall-normal velocity with increasing SFRJ combustor operating pressure. Consistent with the flame thinning and sharpening of gradients observed with increasing pressure in one-dimensional flames, the gradients at the lip edge increase nearly monotonically with increasing pressure. The temperature, CO mass fraction, and CO_2 mass fraction gradient rise is consistent with the decreasing standoff of the flame with increasing temperature observed in the instantaneous profiles. Although there is a clear trend of instantaneous temperature with increasing pressure, the average temperature lipline profiles show overall agreement in peak temperature prediction.

The density along the combustor lipline shows significant differences in the density gradients at the lipline between the pressure cases. Qualitative agreement is seen in the density variation along the lipline between the $p = 270$ kPa and 540 kPa cases, in which there is a sharp rise in density gradient followed by a sharp decline. A much weaker variation is observed in the $p = 135$ kPa case and mostly no variation is observed in the $p = 67.5$ kPa case. This further indicates evidence of the $p = 67.5$ kPa approaching the flammability limits as a result of pressure-dependent chemical kinetics. The $p = 67.5$ kPa shows a much lower axial velocity along the lipline profile indicating less combustion-induced expansion in comparison to other cases. The axial velocity profiles show excellent agreement between the $p = 135$ kPa, 270 kPa, and 540 kPa cases, as away from flammability limits the expansion from combustion is pressure-insensitive. Wall-normal velocity profiles show similar magnitude of variation along the lipline, with steeper gradients observed with increasing pressure.

The prediction of CO and CO_2 mass fraction show good trend-wise agreement with increasing pressure. The gradient of CO_2 mass fraction at the backward facing step increases considerably with increasing pressure, indicating that the reaction progress increases with increasing pressure. There is a slight trend towards increasing CO to CO_2 oxidation with increasing pressure, evidenced by monotonic trends below roughly $x = 0.15$ m. Pressure-dependent reactions in the six-step mechanism favor the production of fewer moles of gas at higher pressures, further driving the CO to CO_2 oxidation.

Figure 19 shows line averages of the grain outflow temperature, density, CO mass fraction, and CO_2 mass fractions, sampled at $x = 0.3$ m. As expected, the wall normal gradients increase monotonically with increasing pressure necessitating the higher resolution at the grain wall surface. The average temperature profiles show qualitative agreement

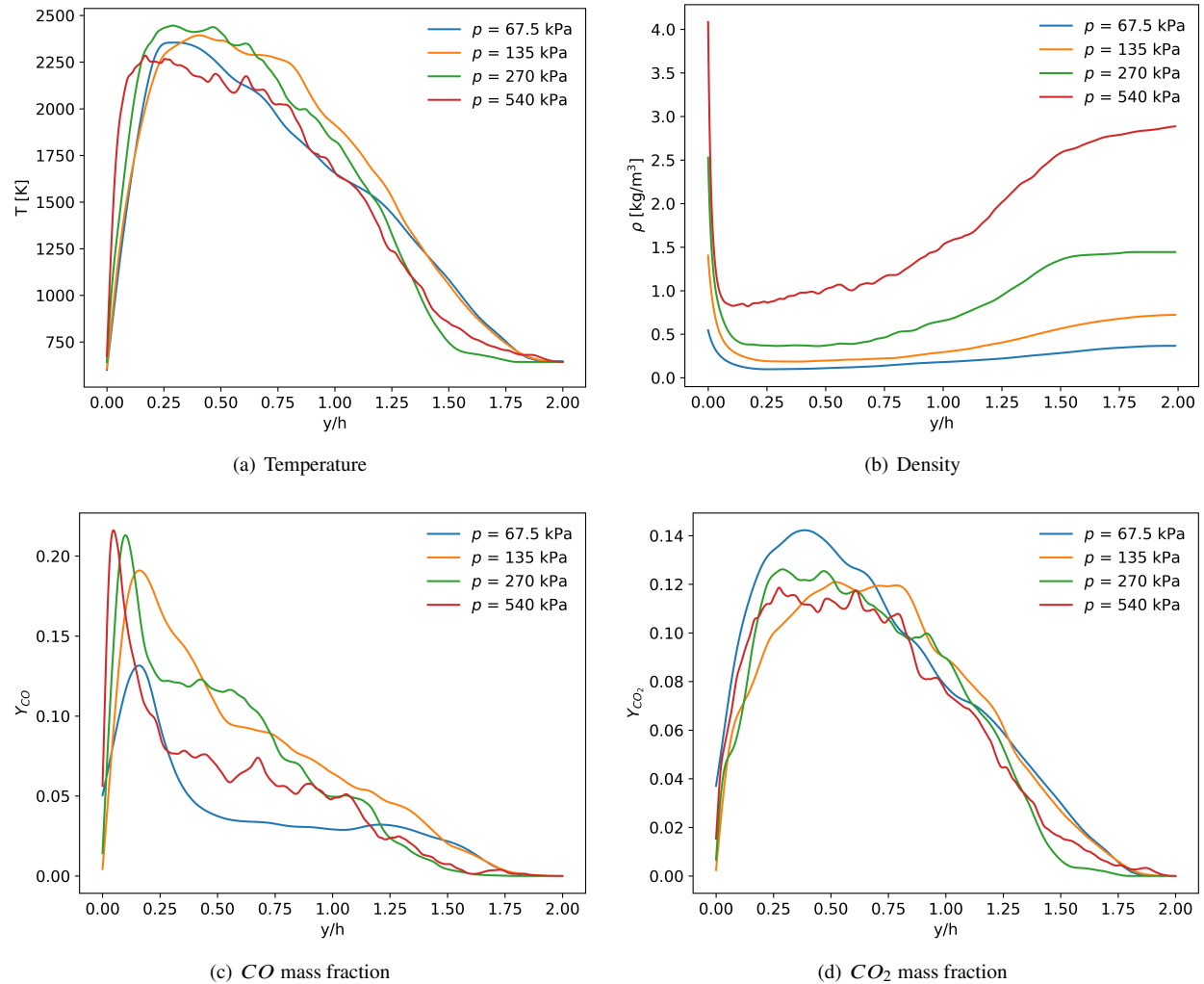


Figure 19 Comparison of average grain outflow profiles, $x = 0.3$, with increasing operating pressure

in shape and quantitative agreement in magnitude, with the peak temperature moving closer to the grain wall surface at $y/h = 0$. Density increases near the wall due to the high density and high mass fraction of C_4H_6 blowing in from the pyrolyzing surface. Profiles of CO and CO_2 mass fraction show good agreement in shape and magnitude, showing an insensitivity to peak CO and CO_2 mass fraction for the $p = 135$ kPa, 270 kPa, and 540 kPa. Table 6 shows the integrated average of temperature and CO , CO_2 , H_2O , and O_2 mass fraction for the outflow profile. Integration was performed from $y/h = 0$ to $y/h = 2$; this choice of wall normal extent mitigates the dampening of results from the largely non-reacting portion of the flowfield. A percent difference is calculated with respect to the $p = 540$ kPa case to assess the global prediction of combustor performance with increasing pressure. Results demonstrate that the integrated combustor performance is largely insensitive to combustor performance, with temperature prediction within 7% for all simulations. The prediction of integrated CO_2 and H_2O show particularly strong agreement maximum deviation of 16% across simulations. Overall agreement with $p = 540$ kPa improves with increasing pressure.

Table 6 Comparison of average integrated grain outflow prediction with percent difference relative to $p = 540$ kPa

Parameter	$p = 67.5 \text{ kPa}$	$p = 135 \text{ kPa}$	$p = 270 \text{ kPa}$	$p = 540 \text{ kPa}$
T [K]	1536.1665 (1.147%)	1625.1813 (7.008 %)	1566.8295 (3.166%)	1518.7457
Y_{CO_2}	0.0735 (16.514%)	0.0675 (7.033%)	0.0641 (1.639 %)	0.0631
Y_{CO}	0.0372 (-24.671 %)	0.0693 (40.301 %)	0.0624 (26.347%)	0.0494
Y_{H_2O}	0.0369 (10.783%)	0.0385 (15.594%)	0.0354 (6.368%)	0.0333
Y_{O_2}	0.1282 (-1.796%)	0.1073 (-17.798 %)	0.1175 (-9.998 %)	0.1305

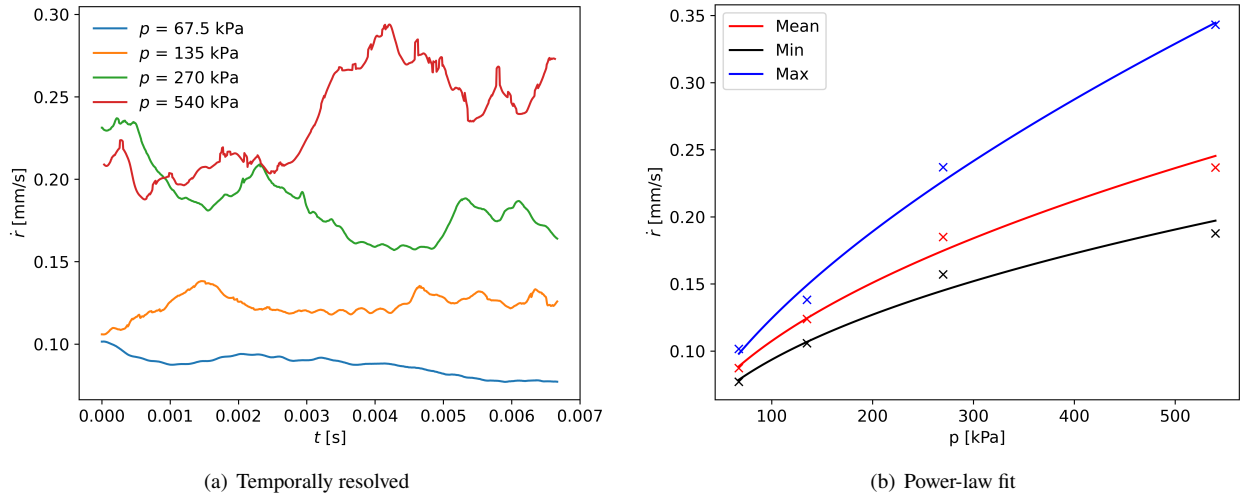


Figure 20 (a) Time history of spatially-averaged regression rate and (b) power law fit with increasing operating pressure

Of most importance to SFRJ performance is the prediction of regression rate, which sets the overall fuel/air ratio of the combustor. Figure 20a shows the transient average regression rate within the combustor with increasing pressure. The transient regression rate across the fuel surface shows an increase in regression rate magnitude with increasing pressure. There is some overlap between the $p = 67.5$ kPa and 135 kPa cases and the $p = 270$ kPa and $p = 540$ kPa cases, due to a wide variance in the time-resolved regression rate [21]. Utilizing Equation 3, power law fits for the mean, minimum, and maximum regression rates are found based on the pressure-scaling simulations. Due to the secondary influence of pressure on density, the power law fits are for the combined pressure and mass flux effects, $\alpha + \beta$. Figure 20b shows the power law fits for the mean, minimum, and maximum spatially averaged regression rates. The corresponding power law fits for each curve is given in Table 7. Uncertainties in power law fits are provided, based on the calculated variances. The results show that the combined effect of pressure and mass flux scale at approximately $p^{0.5}$ range. This result, for the model rectangular burner geometry, is slightly lower than the published empirical fits for axisymmetric SFRJ combustors [26]. However, this result agrees extremely well with the results from the one-dimensional solid fuel counterflow simulations, showing an $\alpha + \beta = 0.48$.

Table 7 1D Power Law Fits for SFRJ Simulations

Parameter	a	$\alpha + \beta$	R^2
Mean regression rate [mm/s]	$3.822\text{e-}04 \pm 1.513\text{e-}04$	0.4898 ± 0.032	0.9958
Min regression rate [mm/s]	$5.766\text{e-}04 \pm 3.211\text{e-}04$	0.442 ± 0.043	0.9904
Max regression rate [mm/s]	$1.177\text{e-}04 \pm 6.482\text{e-}05$	0.6047 ± 0.044	0.9949

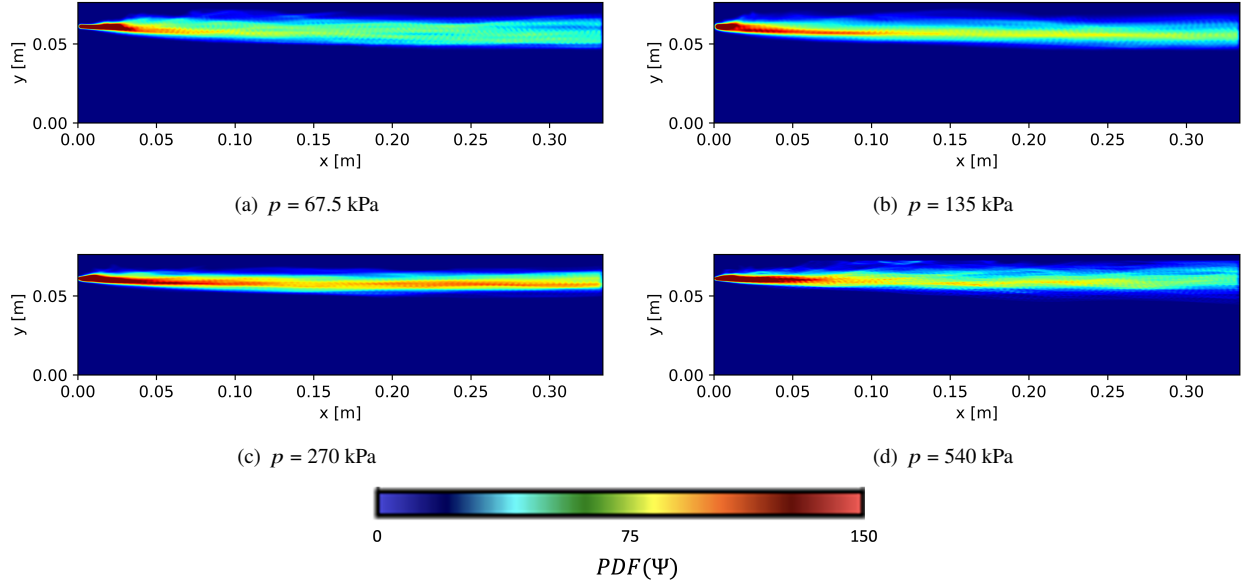


Figure 21 Comparison of flame brush PDF computed for various operating pressures

The leading flame edge was detected for each instantaneous sample and a probability density function (PDF) of flame edge, $PDF(\Psi)$ is constructed at each axial location, where the probability of finding the flame edge within a given radius at an axial station is given by [21]:

$$p(\Psi \in [0, y_{max}]) = \int_0^{y_{max}} PDF(\Psi) dy \quad (30)$$

The result is compiled in Figure 21, which compares the local axial PDF of flame edge with increasing pressure. The flame brush at operating pressure $p = 67.5$ kPa shows the largest width along the axial direction as a consequence of unsteady dynamics resulting from finite-rate Damköhler number effects. The excellent agreement between $p = 135$ kPa, 270 kPa, and 540 kPa highlights an insensitivity on flame dynamics with increasing pressure. All cases show the smallest flame brush width near the leading edge of the backward facing step, where the flame is anchored. The $p = 67.5$ kPa case sees the flame brush width quickly increase, whereas the higher pressure cases see higher stability near the attachment point. Using the instantaneous detected edge, curvature of the flame edge was calculated following the procedure by Jenkins et al. [60, 61]. A surface normal is calculated based on the local gradient in temperature along the

detected edge:

$$N_T = -\frac{\nabla T}{|\nabla T|} \quad (31)$$

where N_T is the local surface normal calculated based on the local temperature gradients. The flame curvature was obtained locally by calculating the gradient of the flame surface normal vector with respect to a rotated coordinate system:

$$\mathbf{h}_{ij} = \frac{\partial N_{T,i}}{\partial x_j} \quad (32)$$

where x_j is oriented perpendicular to the principal direction of the local surface normal. Figure 22 shows a comparison of the calculated flame curvature PDF with increasing operating pressure. The PDF shows the curvature of the leading edge is largely centered around 0 curvature, with tails showing regions of very high curvature. Table 8 summarizes the results of the curvature calculations, in which there is a monotonic increase in maximum flame curvature with increasing operating pressure, consistent with the smaller flame structures observed in the higher pressure. The average flame curvature is in the negative direction, as flame roll up is biased in favor of rolling inwards towards the fuel grain surface. There is a similarity in the average and standard deviation of flame curvature for the $p = 67.5$ kPa and 135 kPa simulations, likely due to the added flame instability near the blowoff condition. Overall, the PDF tails extend further outward for the high pressure cases as a result of local regions of high flame curvature. For non-premixed flames, capturing large curvature effects are important for capturing ignition characteristics and preferential diffusion effects [62].

In comparison to expected one-dimensional solid fuel counterflow theory, pressure scaling has the intended consequence of sharpening gradients, increasing flame curvature, and increasing peak instantaneous temperature. These scaling effects are of secondary consequence to performance, in which the overall combustion efficiency and integrated profiles mostly show insensitivity to increasing pressure. Results of $p = 135$ kPa, 270 kPa, and 540 kPa situations show much better agreement than the $p = 67.5$ kPa case, highlighting that the approach of flammability limits modifies the predicted flame structure considerably. Monotonic increase in regression rate is seen as predicted, with a power law dependence predicted similar to the one-dimensional theory. Deviation can be attributed to comparison to axisymmetric configurations and the non-inclusion of radiation effects. In axisymmetric configurations, the view factor is unity and large changes in regression rate with opacifying agents is observed [63, 64]. The observed monotonic trends highlight the first-order effects of increasing operating pressure, allowing for a suitable mapping of combustor operation across a range of altitudes and flight conditions. Overall insensitivity to the flame brush PDF, flame temperature, and exhaust composition illustrates the self-throttling behavior of the coupled regression and unsteady flame dynamics.

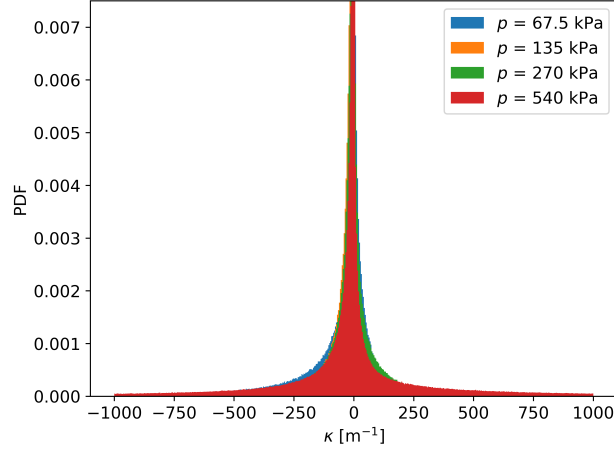


Figure 22 Comparison of flame curvature PDF with increasing operating pressure conditions.

Table 8 Summary comparison of flame curvature metrics with increasing operating pressure

Case	p [kPa]	$\mu(\kappa)$ [m^{-1}]	$\sigma(\kappa)$ [m^{-1}]	$\max(\kappa)$ [m^{-1}]
0.125	67.5	-64.13	1082.38	14773
0.25	135	-44.19	969.99	14778
0.50	270	-53.883	1703.45	35380
1.0	540	-121.03	2877.20	72350.82

D. Experimental Comparison

The results of the pressure scaling simulations are used to compare with high-fidelity experimental measurements made by Senior et al. [50]. Experimental temperature and flame surface probability (FSP) measurements were conducted using coherent anti-Stokes Raman scattering (CARS) and OH planar laser-induced fluorescence imaging (OH-PLIF), respectively. The experimental FSP (analogous to numerical flame brush, $PDF(\Psi)$) was measured by extracting the flame edge from OH-PLIF imaging. Notably, in Ref. [50], the operating conditions to the model SFRJ are slightly different, $T_{in} = 668$ K and $P_{in} = 573$ kPa, than the numerical simulations, $T_{in} = 650$ K and $P_{in} = 540$ kPa.

Figure 23 shows a comparison of experimental and numerical flame temperature at several slices of increasing axial distance along the solid fuel grain. The numerical flame temperature is considerably higher than experiment, with peak flame temperature approximately 500 K higher than experiment across the axial slices. There is some agreement in overall profile distribution; however, numerical temperature profiles have a much higher thermal gradient near the wall in comparison to experiment. The differences in temperature may be attributed to a lack of dimensionality, non-inclusion of radiation or soot effects, and the use of reduced kinetics modeling. There is a clear need for more advanced SFRJ modeling approaches, which will be enabled in part through the cost reduction strategies outlined in this paper.

Figure 24 shows a comparison of experimental FSP and numerical flame brush PDF ($PDF(\Psi)$) at several slices of

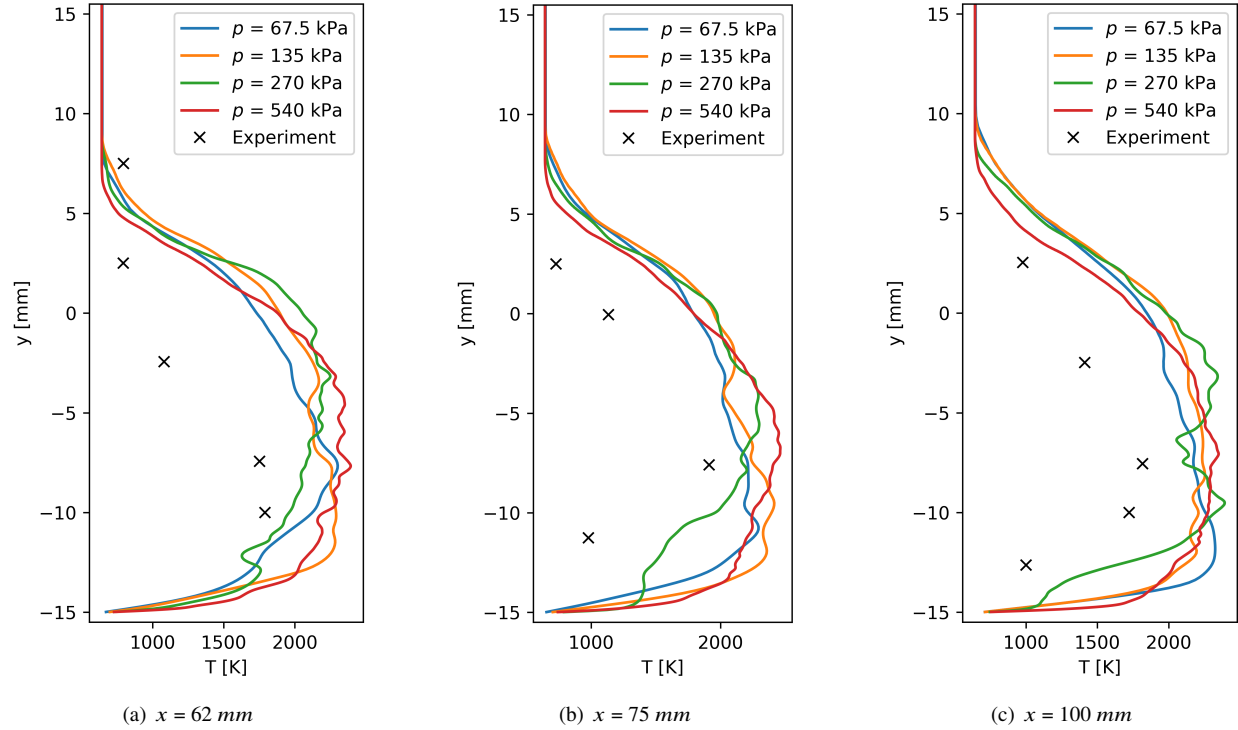


Figure 23 Comparison of experimental (Senior et al. [50]) and numerical flame temperature

increasing axial distance along the solid fuel grain. The numerical flame brush is measured differently than experimental FSP by using a low-temperature (800 K) isotherm to detect the leading edge. This is required as the inclusion of OH in the reduced six-step mechanism is only for mitigating peak temperature through dissociation (Equation R8); therefore, it is not an accurate predictor of physical OH distribution [21]. To compare between the different flame brush metrics, a translated y -coordinate, $\tilde{y}$, is introduced that centers the associated PDFs at $\tilde{y} = 0$. The experimental and numerical flame brush probabilities show very good agreement across the range of sampled axial locations. As demonstrated in the previous Subsection, the flame brush is fairly consistent with increasing combustor pressure in numerical simulations. The flame brush widens with increasing x , with the numerical simulations bounding the experimental flame brush along the majority of $\tilde{y}$. Above $\tilde{y} \approx 5$, the experimental and numerical flame brushes start to diverge, with numerical simulations showing a sharper drop in probability. This may be attributed to the lack of unsteady flow-chemistry interactions of the reduced global mechanism, leading to less flame protrusion above the leading edge in comparison to more detailed chemical kinetics [21].

V. Conclusion

This work provides an understanding of the unsteady dynamics within an SFRJ combustor operating at a range of altitude conditions, while outlining viable strategies for reducing the computational cost of physically-accurate

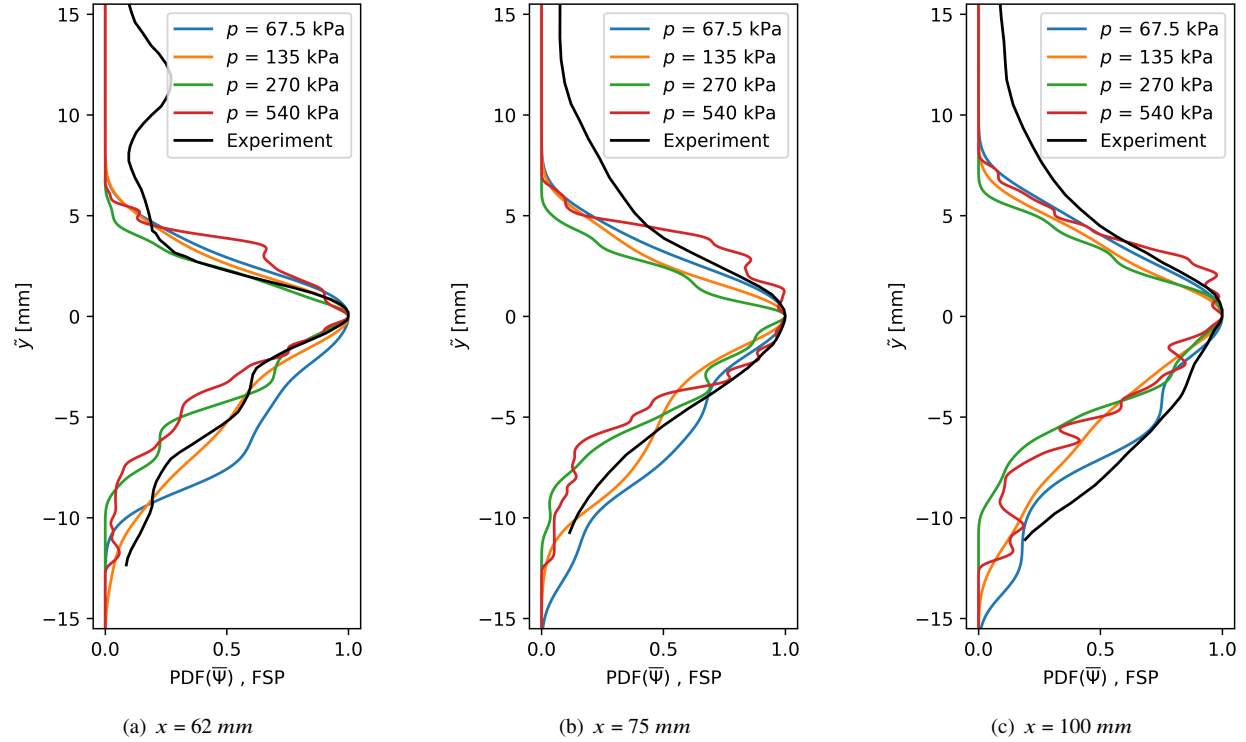


Figure 24 Comparison of experimental flame surface probability (FSP) [50] and numerical flame brush PDF ($PDF(\Psi)$). $\tilde{y}$: translated y -coordinate centered at the maximum value.

numerical simulations. Two-dimensional parametric LES simulations are performed on a model planar SFRJ combustor using interfacial solid fuel pyrolysis modeling of a model HTPB grain.

Numerical simulations were first performed to outline strategies for reducing computational cost. Comparisons between combustor models utilizing subsonic and supersonic outflow show minimal effects on the predicted combustion behavior. Utilizing a subsonic outflow resulted in half the computational expense by minimizing the total number of elements and increasing the minimum global time step. A mesh refinement study was performed at the lowest operating pressure, $p = 67.5$ kPa, to assess the relationship between resolution requirements at the inflow and the solid-fuel interface boundary. A decrease in surface resolution results from the combined effects of flow expansion, increase in viscosity with combustion temperature rise, and blowing velocity from the fuel surface. The results of the mesh refinement study demonstrated that a mesh scaling factor of $\sigma = 5$ was able to accurately reproduce the average SFRJ regression rate, reattachment point, and instantaneous and average flame contours. SPOD and POD are also performed, confirming that the mesh resolution is able to reproduce the unsteady dynamics within the model SFRJ.

Using the resolution requirements found in the mesh refinement study, 2D SFRJ LES were conducted with increasing pressures, $p = 67.5$ kPa, 135 kPa, 270 kPa, and 540 kPa. Calculation of local y^+ values along the fuel grain surface showed that the mesh scaling factor held with increasing Reynolds number, establishing a methodology for determining mesh resolution apriori to larger calculations. Instantaneous contours, average line plots, and time-resolved regression

rates affirmed one-dimensional theory in which sharper gradients, increased regression rate, and increased maximum flame temperature were observed with increasing pressure. Calculation of outflow exhaust composition and temperature showed relative insensitivity to combustor performance with increasing pressure, highlighting the self-throttling nature of the SFRJ engine. Deviations in flame-holding characteristics and predicted flame brush at $p = 67.5$ kPa indicates that combustor dynamics change considerably near blowoff conditions. Higher pressure conditions of $p = 135$ kPa, 270 kPa, and 540 kPa showed monotonic increase in flame curvature and relative insensitivity to predicted flame brush. The cumulative results indicate that away from blowoff conditions, capturing the underlying combustion physics at higher altitudes, amenable to lower computational expense, leads to significant insight into lower altitude operation.

Numerical simulations with increasing pressure were then compared to experimental temperature and flame brush measurement. The simulations overpredict the temperature considerably, owing to limitations in model chemistry, dimensionality, and inclusion of radiation and soot effects. The flame brush agrees very well between experiment and numerical simulation, with a slow widening of flame brush observed with increasing axial distance.

While this paper takes a step towards understanding SFRJ combustion, the complex multiphysics phenomena is far from understood. LES offers the ability to directly probe the combustion physics, demonstrated to be more accurate and drive further insight compared to RANS approaches [12, 19–21]. There remains a clear need for advancing current high-fidelity SFRJ simulation capability, which will include modeling full three-dimensionality, detailed combustion chemistry, soot, and radiation. This is highlighted by the overprediction of temperature by the present model as a consequence of reduced-order modeling. Developing robust models for these phenomena are also necessary for enabling further breakthroughs in combustor design with fuels containing metal additives for greater energy densities. Validation of numerical models against detailed experiments (enabled through optical access) is vital to extending simulation capability to real-world geometries that include aft-mixing, air bypass, and advanced fuel geometries. Ultimately, these advances, coupled with high-fidelity LES, will enable full-scale system prediction, shortening the SFRJ design cycle and enabling widespread operational deployment.

Appendix: One-Dimensional Counterflow Formulation

The governing equations of a quasi-one-dimensional steady-state axisymmetric stagnation flow solver are given below:

$$\frac{\partial \rho v_z}{\partial z} + 2\rho V = 0 \quad (33)$$

$$\rho v_z \frac{\partial Y_i}{\partial z} = -\frac{\partial j_i}{\partial z} + W_i \omega_i \quad (34)$$

$$\rho v_z \frac{\partial V}{\partial z} + \rho V^2 = -\Lambda + \frac{\partial}{\partial z} \left(\mu \frac{\partial V}{\partial z} \right) \quad (35)$$

$$\rho c_p v_z \frac{\partial T}{\partial z} = \frac{\partial}{\partial z} \left(\lambda_T \frac{\partial T}{\partial z} \right) - \sum_i j_i c_{p,i} \frac{\partial T}{\partial z} - \sum_i h_i W_i \omega_i \quad (36)$$

where z is axial direction coordinate, ρ is the fluid density, v_z is the axial velocity, $V = v_r/r$ is the scaled radial velocity or spread rate, Y_i is the mass fraction of the i th species, j_i is the diffusive mass flux of the i th species, Λ is the pressure eigenvalue (independent of z), T is the temperature, λ_T is the thermal conductivity, μ is the dynamic viscosity, $c_{p,i}$ is the constant pressure heat capacity, h_i is the enthalpy, W_i is the molecular weight, and ω_i is the source term of the i th species. Grid refinement was used in this study and further details on the solution procedure and convergence parameters for the gas-phase solver can be seen in Refs. [28, 40].

A fully coupled solid-fuel boundary condition is used to define the fuel mass flux, pyrolysis gases, and surface temperature of the HTPB fuel grain [21, 33]. The conservation of mass, species and energy at the boundary (identical to LES Equations 21-24) are defined as :

$$\dot{m}_s'' = \rho v = \rho_s \dot{r} \quad (37)$$

$$\dot{m}_s'' = \rho v Y_{sg} - \rho_g D_F \left(\frac{\partial Y_{sg}}{\partial z} \right)_f \quad (38)$$

$$0 = \rho v Y_i - \rho D_i \left(\frac{\partial Y_i}{\partial z} \right)_f \quad (39)$$

$$\dot{m}_s'' L_v = -\lambda_f \left(\frac{\partial T}{\partial z} \right)_f \quad (40)$$

where the subscripts 's' and 'i' refer to the gas-phase pyrolysis product species and the other non-participating species, respectively. The right-hand side represents the gas-phase fluid, denoted by subscript 'f', and the left-hand side represents the solid-phase. Derivatives were computed in the along the axial direction, ' $\hat{z}$ ', and L is the temperature-dependent effective latent heat of vaporization of the solid species. The surface blowing velocity is $v = \dot{m}_s''/\rho$, where ρ is the density of the gas in the pyrolysis zone. The regression rate, $\dot{r}$, is modeled based on an Arrhenius rate form:

$$\dot{r} = l_r A_p e^{-E_p/RT_I}, \quad (41)$$

where l_r is the reaction thickness of the pyrolysis zone, and T_I is the temperature at the fluid-solid interface. A_p and E_p

are empirical fits for the pre-exponential factor and activation energy, respectively, given in Table 9. The solid fuel is composed entirely of a hypothetical polymer with properties similar to hydroxyl-terminated polybutadiene (HTPB) that is converted entirely to the gaseous hydrocarbon, butadiene (C_4H_6). The thermophysical properties of the model HTPB solid fuel are given from Arisawa and Brill [45, 46] and provided in Table 10. Equations (38-41) then define a set of n_s + 2 nonlinear equations, which solves for the species mass fractions, mass flux, and interfacial temperature. The set of equations are solved using an iterative Gauss-Newton method. The gas-phase solution is subsequently re-calculated until convergence is met between both phases.

Table 9 HTPB pyrolysis parameters [45, 46]

A_p	E_p	l_r
$4.0889 \times 10^4 \text{ s}^{-1}$	$5.65 \times 10^7 \text{ J/kmol}$	$6 \times 10^{-5} \text{ m}$

Table 10 Thermophysical properties of model HTPB [45, 46]

Density	Specific Heat	Thermal Conductivity	Heat of Formation
900 kg/m^3	1100 J/kg K	0.235 W/m K	54.4 kJ/kg

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References

- [1] Netzer, D. W., "Modeling Solid-Fuel Ramjet Combustion," *Journal of Spacecraft and Rockets*, Vol. 14, 1977, pp. 762–766. <https://doi.org/10.2514/3.27994>.
- [2] Stevenson, C. A., and Netzer, D. W., "Primitive-Variable Model Applications to Solid-Fuel Ramjet Combustion," *Journal of Spacecraft and Rockets*, Vol. 18, 1981, pp. 89–94. <https://doi.org/10.2514/3.28050>.
- [3] Metochianakis, M. E., and Netzer, D. W., "Modeling Solid-Fuel Ramjet Combustion, Including Radiation to the Fuel Surface," *Journal of Spacecraft and Rockets*, Vol. 20, 1983, pp. 405–406. <https://doi.org/10.2514/3.25617>.
- [4] Milshtein, T., and Netzer, D. W., "Three-Dimensional, Primitive-Variable Model for Solid-Fuel Ramjet Combustion," *Journal of Spacecraft and Rockets*, Vol. 23, 1986, pp. 113–117. <https://doi.org/10.2514/3.25792>.

- [5] Elands, P. J., Korting, P. A., Wijchers, T., and Dijkstra, F., "Comparison of Combustion Experiments and Theory in Polyethylene Solid Fuel Ramjets," *Journal of Propulsion and Power*, Vol. 6, 1990, pp. 732–739. <https://doi.org/10.2514/3.23279>.
- [6] Liou, T. M., Lien, W. Y., and Hwang, P. W., "Flammability Limits and Probability Density Functions in Simulated Solid-Fuel Ramjet Combustors," *Journal of Propulsion and Power*, Vol. 13, 1997, pp. 643–650. <https://doi.org/10.2514/2.5229>.
- [7] Musa, O., Xiong, C., Changsheng, Z., and Li, W., "Effect of Inlet Conditions on Swirling Turbulent Reacting Flows in a Solid Fuel Ramjet Engine," *Applied Thermal Engineering*, Vol. 113, 2017, pp. 186–207. <https://doi.org/10.1016/j.applthermaleng.2016.11.023>.
- [8] Gong, L., Chen, X., and Musa, O., "Modeling of the Turbulent Combustion in Solid-fuel Ramjet," *Energy Procedia*, Vol. 141, 2017, pp. 267–272. <https://doi.org/10.1016/j.egypro.2017.11.104>.
- [9] Li, W., Zhao, D., Chen, X., Zhu, L., and Ni, S., "Numerical Investigation of Inlet Thermodynamic Conditions on Solid Fuel Ramjet Performances," *International Journal of Aerospace Engineering*, Vol. 2021, 2021. <https://doi.org/10.1155/2021/8868288>.
- [10] Bianchi, D., Nasuti, F., and Onofri, M., "Radius of Curvature Effects on Throat Thermochemical Erosion in Solid Rocket Motors," *Journal of Spacecraft and Rockets*, Vol. 52, 2015, pp. 320–330. <https://doi.org/10.2514/1.A32944>.
- [11] Rashkovskiy, S., Yakush, S., and Baranov, A., "Combustion Stability in a Solid-Fuel Ramjet Engine," *Journal of Physics: Conference Series*, Vol. 1009, 2018, p. 012032. <https://doi.org/10.1088/1742-6596/1009/1/012032>.
- [12] DeBoskey, R., Goodwin, G. B., Kessler, D. A., and Narayanaswamy, V., "Comparison of Turbulence and Chemistry Closures in a Solid-Fuel Ramjet Combustor," *Journal of Propulsion and Power*, 2025. <https://doi.org/10.2514/1.B39977>.
- [13] Bianchi, D., Leccese, G., Nasuti, F., Onofri, M., and Carmicino, C., "Modeling of High Density Polyethylene Regression Rate in the Simulation of Hybrid Rocket Flowfields," *Aerospace*, Vol. 6, 2019, p. 88. <https://doi.org/10.3390/aerospace6080088>.
- [14] Coronetti, A., and Sirignano, W. A., "Numerical Analysis of Hybrid Rocket Combustion," *Journal of Propulsion and Power*, Vol. 29, 2013, pp. 371–384. <https://doi.org/10.2514/1.B34760>.
- [15] Liou, T. M., Lien, W. Y., and Hwang, P. W., "Large-Eddy Simulations of Turbulent Reacting Flows in a Chamber with Gaseous Ethylene Injecting through the Porous Wall," *Combustion and Flame*, Vol. 99, 1994, pp. 591–600. [https://doi.org/10.1016/0010-2180\(94\)90052-3](https://doi.org/10.1016/0010-2180(94)90052-3).
- [16] Morínigo, J. A., and Hermida-Quesada, J., "Evaluation of Reduced-Order Kinetic Models for HTPB-Oxygen Combustion using LES," *Aerospace Science and Technology*, Vol. 58, 2016, pp. 358–368. <https://doi.org/10.1016/j.ast.2016.08.027>.
- [17] Park, N. S., and Ko, S. C., "Large Eddy Simulation of Turbulent Premixed Combustion Flows over Backward Facing Step," *Journal of Mechanical Science and Technology*, Vol. 25, 2011, pp. 713–719. <https://doi.org/10.1007/s12206-011-0106-8>.
- [18] Nemitallah, M. A., Kewlani, G., Hong, S., Shanbhogue, S. J., Habib, M. A., and Ghoniem, A. F., "Investigation of a Turbulent Premixed Combustion Flame in a Backward-Facing Step Combustor; Effect of Equivalence Ratio," *Energy*, Vol. 95, 2016, pp. 211–222. <https://doi.org/10.1016/j.energy.2015.12.010>.

- [19] Gallegos, D., Pace, H., Arnold, C., Massa, L., and Young, G., “Regression and Flame Structure in Cavity Flameholding Solid-Fuel Ramjet Fuel Grains,” *Journal of Propulsion and Power*, Vol. 39, No. 6, 2023, pp. 800–810. <https://doi.org/10.2514/1.B39139>.
- [20] Pace, H., Arnold, C., Gallegos, D., Young, G., and Massa, L., “Acoustics in Flameholding Solid-Fuel-Ramjet Fuel Grains,” *AIAA Journal*, Vol. 61, No. 10, 2023, pp. 4579–4597. <https://doi.org/10.2514/1.J062832>.
- [21] DeBoskey, R., Kessler, D. A., Bojko, B., Johnson, R. F., Kercher, A., Ching, E., and Narayanaswamy, V., “Development and Evaluation of Reduced-Order Kinetics Models for 1,3-Butadiene-Air Combustion,” *Journal of Propulsion and Power*, 2025. <https://doi.org/10.2514/1.B39768>.
- [22] Veraar, R. G., Oosthuisen, R., and Andersson, K., “Ramjet Propulsion for Projectiles - An Overview of Worldwide Achievements and Future Opportunities,” *International Journal of Energetic Materials and Chemical Propulsion*, Vol. 21, No. 5, 2022, pp. 1–61. <https://doi.org/10.1615/IntJEnergeticMaterialsChemProp.2022038741>.
- [23] DeBoskey, R., Sahoo, A., Narayanaswamy, V., and Kessler, D., “Demonstration of Fuel PLIF in a Model Solid Fuel Ramjet Combustor,” *Applied Optics*, Vol. 64, 2025, pp. 6966–6974. <https://doi.org/10.1364/AO.565558>.
- [24] Veraar, R. G., and Wieling, W. P., “Sustained Combustion Limits of a Central Dump Solid Fuel Ramjet Combustor at High Altitude Operational Conditions,” AIAA 2018-4449, July 2018. <https://doi.org/10.2514/6.2018-4449>.
- [25] Schulte, G., “Fuel Regression and Flame Stabilization Studies of Solid-Fuel Ramjets,” *Journal of Propulsion and Power*, Vol. 2, 1986, pp. 301–304. <https://doi.org/10.2514/3.22886>.
- [26] LeMoyne, R., “Fundamental Analysis of a Solid Fuel Ramjet,” AIAA 2012-5904, September 2012. <https://doi.org/10.2514/6.2012-5904>.
- [27] Hankins, W. J., “Assessment of Performance Parameter Uncertainty of a Solid Fuel Ramjet,” Master’s Thesis, Dept. of Mechanical and Aerospace Engineering, Univ. of Alabama in Huntsville, Huntsville, AL, 1983.
- [28] Goodwin, D. G., Moffat, H. K., Schoegl, I., Speth, R. L., and Weber, B. W., “Cantera: An Object-oriented Software Toolkit for Chemical Kinetics, Thermodynamics, and Transport Processes,” <https://www.cantera.org>, 2022. <https://doi.org/10.5281/zenodo.6387882>, version 2.6.0.
- [29] Geipel, C. M., Bojko, B. T., Pfützner, C. J., Fisher, B. T., and Johnson, R. F., “Regression of Solid Polymer Fuel Strands in Opposed-Flow Combustion with Gaseous Oxidizer,” *Proceedings of the Combustion Institute*, Vol. 39, No. 3, 2023, pp. 3389–3399. <https://doi.org/10.1016/j.proci.2022.07.124>.
- [30] Bojko, B. T., Geipel, C. M., Fisher, B. T., and Kessler, D. A., “Numerical sensitivity analysis of HTPB counterflow combustion using neural networks,” *Combustion and Flame*, Vol. 271, 2025, p. 113829. <https://doi.org/10.1016/j.combustflame.2024.113829>.
- [31] Laskin, A., Wang, H., and Law, C. K., “Detailed Kinetic Modeling of 1, 3-Butadiene Oxidation at High Temperatures,” *International Journal of Chemical Kinetics*, Vol. 32, No. 10, 2000, pp. 589–614. [https://doi.org/10.1002/1097-4601\(2000\)32:10<589::AID-KIN2>3.0.CO;2-U](https://doi.org/10.1002/1097-4601(2000)32:10<589::AID-KIN2>3.0.CO;2-U).

- [32] Senior, W. C., Gejji, R. M., and Slabaugh, C. D., “Flame Dynamics in an Optically Accessible Solid Fuel Ramjet Combustor,” *Journal of Propulsion and Power*, Vol. 39, No. 5, 2023, pp. 718–727. <https://doi.org/10.2514/1.B39078>.
- [33] DeBoskey, R., Geipel, C., Kessler, D., Bojko, B., Fisher, B., Johnson, R. F., and Narayanaswamy, V., “Numerical and Experimental Investigation of Flame Dynamics in Opposed-Flow Solid Fuel Burner,” *Combustion and Flame*, Vol. 273, 2025, p. 113960. <https://doi.org/10.1016/j.combustflame.2024.113960>.
- [34] Vancoillie, J., Demuynck, J., Galle, J., Verhelst, S., and Oijen, J. A. V., “A Laminar Burning Velocity and Flame Thickness Correlation for Ethanol-Air Mixtures Valid at Spark-Ignition Engine Conditions,” *Fuel*, Vol. 102, 2012, pp. 460–469. <https://doi.org/10.1016/j.fuel.2012.05.022>.
- [35] McBride, B. J., Gordon, S., and Reno, M. A., “Coefficients for Calculating Thermodynamic and Transport Properties of Individual Species,” Tech. Rep. NASA-TM-4513, NASA Glenn Research Center, Cleveland Ohio, 1993.
- [36] McBride, B. J., Zehe, M. J., and Gordon, S., “NASA Glenn Coefficients for Calculating Thermodynamic Properties of Individual Species,” Tech. Rep. NASA/TP-2002-211556, NASA Glenn Research Center, Cleveland Ohio, 2002.
- [37] Coffee, T., and Heimerl, J., “Transport Algorithms for Premixed, Laminar Steady-State Flames,” *Combustion and Flame*, Vol. 43, 1981, pp. 273 – 289. [https://doi.org/https://doi.org/10.1016/0010-2180\(81\)90027-4](https://doi.org/https://doi.org/10.1016/0010-2180(81)90027-4).
- [38] Houim, R., and Kuo, K., “A Low-Dissipation and Time-Accurate Method for Compressible Multi-Component Flow with Variable Specific Heat Ratios,” *Journal of Computational Physics*, Vol. 230, No. 23, 2011, pp. 8527 – 8553. <https://doi.org/https://doi.org/10.1016/j.jcp.2011.07.031>.
- [39] Kee, R. J., Miller, J. A., Evans, G. H., and Dixon-Lewis, G., “A Computational Model of the Structure and Extinction of Strained, Opposed Flow, Premixed Methane-Air Flames,” *Symposium (International) on Combustion*, Vol. 22, No. 1, 1989, pp. 1479 – 1494. [https://doi.org/https://doi.org/10.1016/S0082-0784\(89\)80158-4](https://doi.org/https://doi.org/10.1016/S0082-0784(89)80158-4).
- [40] Kee, R. J., Coltrin, M. E., and Glarborg, P., *Chemically Reacting Flow: Theory and Practice*, Wiley, Hoboken, NJ, 2005.
- [41] R. Wilke, C., “A Viscosity Equation for Gas Mixtures,” *Journal of Chemical Physics*, Vol. 18, 1950, pp. 517–519. <https://doi.org/10.1063/1.1747673>.
- [42] Mathur, S., Tondon, P. K., and Saxena, S. C., “Thermal Conductivity of Binary, Ternary and Quaternary Mixtures of Rare Gases,” *Molecular Physics*, Vol. 12, 1967, pp. 569–579. <https://doi.org/10.1080/00268976700100731>.
- [43] Johnson, R. F., and Kercher, A. D., “A Conservative Discontinuous Galerkin Discretization for the Chemically Reacting Navier-Stokes Equations,” *Journal of Computational Physics*, Vol. 423, 2020, p. 109826. <https://doi.org/10.1016/j.jcp.2020.109826>.
- [44] Hartmann, R., and Houston, P., “An Optimal Order Interior Penalty Discontinuous Galerkin Discretization of the Compressible Navier-Stokes Equations,” *Journal of Computational Physics*, Vol. 227, 2008, pp. 9670–9685. <https://doi.org/10.1016/j.jcp.2008.07.015>.

- [45] Arisawa, H., and Brill, T. B., "Flash Pyrolysis of Hydroxyl-Terminated Polybutadiene (HTPB) I: Analysis and Implications of the Gaseous Products," *Combustion and Flame*, Vol. 106, 1996, pp. 131–143. [https://doi.org/10.1016/0010-2180\(96\)00253-2](https://doi.org/10.1016/0010-2180(96)00253-2).
- [46] Arisawa, H., and Brill, T., "Flash Pyrolysis of Hydroxyl-Terminated Polybutadiene (HTPB) II: Implications of the Kinetics to Combustion of Organic Polymers," *Combustion and Flame*, Vol. 106, No. 1-2, 1996, pp. 144–154. [https://doi.org/https://doi.org/10.1016/0010-2180\(95\)00254-5](https://doi.org/https://doi.org/10.1016/0010-2180(95)00254-5).
- [47] Chiaverini, M. J., Harting, G. C., Lu, Y. C., Kuo, K. K., Peretz, A., Jones, H. S., Wygle, B. S., and Arves, J. P., "Pyrolysis Behavior of Hybrid-Rocket Solid Fuels Under Rapid Heating Conditions," *Journal of Propulsion and Power*, Vol. 15, 1999, pp. 888–895. <https://doi.org/10.2514/2.5512>.
- [48] Kee, R. J., Rupley, F. M., and Miller, J. A., "Chemkin-II: A Fortran Chemical Kinetics Package for the Analysis of Gas-Phase Chemical Kinetics," Tech. Rep. SAND-89-8009, Sandia National Lab.(SNL-CA), Livermore, CA (United States), 1989.
- [49] Senior, W. C., Evans, J. V., Gejji, R. M., and Slabaugh, C. D., "Development and Investigation of an Optically Accessible Solid Fuel Ramjet Combustor," AIAA 2021-1019, January 2021. <https://doi.org/10.2514/6.2021-1019>.
- [50] Senior, W. C., Macarie, L., Gai, T., Gejji, R. M., Lucht, R. P., and Slabaugh, C. D., "Temperature, Oxygen Concentration, and Flame Structure Measurements in Solid Fuel Ramjet Combustor," *AIAA Journal*, Vol. 63, No. 8, 2025, pp. 3434–3444. <https://doi.org/10.2514/1.J064744>.
- [51] Ganji, A. R., and Sawyer, R. F., "Experimental Study of the Flowfield of a Two-Dimensional Premixed Turbulent Flame," *AIAA Journal*, Vol. 18, 1980, pp. 817–824. <https://doi.org/10.2514/3.50823>.
- [52] White, F. M., *Fluid Mechanics*, 6th ed., McGraw-Hill Series in Mechanical Engineering, McGraw-Hill, New York, NY, 2009.
- [53] Hanks, R. W., and Ruo, H.-C., "Laminar-Turbulent Transition in Ducts of Rectangular Cross Section," *Industrial & Engineering Chemistry Fundamentals*, Vol. 5, No. 4, 1966, pp. 558–561. <https://doi.org/10.1021/i160020a022>.
- [54] Spazzini, P. G., Iuso, G., Onorato, M., Zurlo, N., and Cicca, G. M. D., "Unsteady Behavior of Back-Facing Step Flow," *Experiments in Fluids*, Vol. 30, 2001, p. 551–561. <https://doi.org/10.1007/s003480000234>.
- [55] Wee, D., Yi, T., Annaswamy, A., and Ghoniem, A. F., "Self-Sustained Oscillations and Vortex Shedding in Backward-Facing Step Flows: Simulation and Linear Instability Analysis," *Physics of Fluids*, Vol. 16, 2004, pp. 3361–3373. <https://doi.org/10.1063/1.1773091>.
- [56] Weiss, J., "A Tutorial on the Proper Orthogonal Decomposition," AIAA 2019-3333, June 2019. <https://doi.org/10.14279/depositonce-8512>.
- [57] Schmidt, O. T., and Colonius, T., "Guide to Spectral Proper Orthogonal Decomposition," *AIAA Journal*, Vol. 58, 2020, pp. 1023–1033. <https://doi.org/10.2514/1.J058809>.

- [58] Zienkiewicz, O. C., and Codina, R., “A General Algorithm for Compressible and Incompressible Flow—Part I. The Split, Characteristic-Based Scheme,” *International Journal for Numerical Methods in Fluids*, Vol. 20, No. 8-9, 1995, pp. 869–885. <https://doi.org/10.1002/fld.1650200812>.
- [59] Zhou, C.-W., Li, Y., Burke, U., Banyon, C., Somers, K. P., Ding, S., Khan, S., Hargis, J. W., Sikes, T., Mathieu, O., Petersen, E. L., AlAbbad, M., Farooq, A., Pan, Y., Zhang, Y., Huang, Z., Lopez, J., Loparo, Z., Vasu, S. S., and Curran, H. J., “An Experimental and Chemical Kinetic Modeling Study of 1,3-Butadiene Combustion: Ignition Delay Time and Laminar Flame Speed Measurements,” *Combustion and Flame*, Vol. 197, 2018, pp. 423–438. <https://doi.org/10.1016/j.combustflame.2018.08.006>.
- [60] Jenkins, K., and Cant, R., “Curvature Effects on Flame Kernels in a Turbulent Environment,” *Proceedings of the Combustion Institute*, Vol. 29, No. 2, 2002, pp. 2023–2029. [https://doi.org/10.1016/S1540-7489\(02\)80247-4](https://doi.org/10.1016/S1540-7489(02)80247-4).
- [61] Gashi, S., Hult, J., Jenkins, K. W., Chakraborty, N., Cant, S., and Kaminski, C. F., “Curvature and Wrinkling of Premixed Flame Kernels—Comparisons of OH PLIF and DNS Data,” *Proceedings of the Combustion Institute*, Vol. 30, No. 1, 2005, pp. 809–817. <https://doi.org/10.1016/j.proci.2004.08.003>.
- [62] Göktolga, M. U., de Goey, L. P. H., and Van Oijen, J. A., “Modeling Curvature Effects in Turbulent Autoigniting Non-Premixed Flames using Tabulated Chemistry,” *Proceedings of the Combustion Institute*, Vol. 38, No. 2, 2021, pp. 2741–2748. <https://doi.org/10.1016/j.proci.2020.07.073>.
- [63] Carmicino, C., Scaramuzzino, F., and Sorge, A. R., “Trade-Off Between Paraffin-Based and Aluminium-Loaded HTPB Fuels to Improve Performance of Hybrid Rocket Fed with N₂O,” *Aerospace Science and Technology*, Vol. 37, 2014, pp. 81–92. <https://doi.org/10.1016/j.ast.2014.05.010>.
- [64] Evans, J. V., Senior, W. C., Gejji, R. M., and Slabaugh, C. D., “Performance of a Solid-Fuel Ramjet Combustor with Bypass Air Addition,” *Journal of Propulsion and Power*, Vol. 39, 2023, pp. 167–175. <https://doi.org/10.2514/1.B38788>.